Geography Division Transformation and Application Modernization (GEOTAM)

Draft Order PWS

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1.0 Introduction and Purpose

The U.S. Census Bureau (USCB) Geography Division (GEO) is procuring application development, transformative technology, communication, and documentation services to modernize existing applications and develop new applications in support of the USCB's Master Address File (MAF)/Topologically Integrated Geographic Encoding and Referencing (TIGER) System (MAF/TIGER), the Geographic Support Program (GSP), Decennial Census, and other ongoing surveys and programs at the USCB.

Services include, but are not limited to:

- Order and Program Management
- IT Project and Program Management
- Application Development and Management Services

1.1 Background

GEO and Decennial Information Technology Division (DITD) develop, modernize, and maintain the MAF/TIGER System comprised of the software applications and databases utilized by GEO to accomplish its mission.

The MAF/TIGER database (MTDB) is a national seamless database integrating spatial and address data, based on Oracle Spatial and the Oracle Topology Data Model. The implementation is in an environment hosted on the Oracle Exadata platform supporting multiple Oracle databases. The DITD develops and maintains the interactive and batch applications that make up the core of the MAF/TIGER System.

The MAF/TIGER System encompasses all address, boundary, and feature data serving as the foundation for data collection, tabulation, and dissemination at the USCB. The System produces twenty percent of the Nation's geospatial data as recognized by the Federal Geographic Data Committee through implementation of the Geospatial Data Act of 2018.

The geographic data from the MAF/TIGER System contribute to the efficient and accurate collection, tabulation, and dissemination of data from the Decennial Census, the American Community Survey (ACS), the Economic Census, as well as annual estimates and surveys, such as the Population Estimates Program (PEP).

GEO collaborates with other divisions in the USCB, including, but not limited to, the American Community Survey Office (ACSO), Decennial Census Management Division (DCMD), DITD, Decennial Statistical Studies Division (DSSD), Economic Directorate, and the Population Division (POP) to execute its mission.

2.0 Scope

This Order provides Order and Program Management, IT Project and Program Management, and Application Development and Management Services support necessary for the MAF/TIGER System and GSP to successfully deliver on USCB and National priorities.

The goals for this Order include:

- Minimizing lifecycle costs and technical debt by modernizing USCB programs' application landscape.
- Fulfilling all requirements on time and within budget within the span of control for the Government and contractor.

- Improving efficiency by automating tasks prevalent within workflows.
- Optimizing application architectures to support large-scale, real-time data processing, enabling USCB to accelerate the speed, improve the quality, and expand the quantity of data analytics used for mission planning and execution.
- Enhancing resilience by implementing industry leading security practices and approaches including but not limited to Zero Trust principles and DevSecOps methodologies.
- Supporting skill development of the USCB's workforce through access to subject matter experts, knowledge transfer, and overall development of a culture of learning and innovation.
- Providing a collaborative effort across Contractors, USCB Federal personnel, and other stakeholders for a shared purpose of driving the success of USCB Programs.

The GEOTAM Order will be issued to the Geospatial BPA under the CenTAM Acquisition framework.

3.0 Task Areas

This section describes the technical and support objectives and services the Contractor will perform. The Contractor shall develop solutions and/or provide the capabilities and support in the following task areas:

- Task Area 1 – Order and Program Management
- Task Area 2 – IT Project and Program Management
- Task Area 3 – Application Development and Management Services

The requirements listed in Task Area 1 – Task Area 3 will be further expounded on in Technical Instructions (TIs).

3.1 Task Area 1 – Order and Program Management

The Contractor shall provide a single, coordinated management framework that links procurement administration and program execution oversight so that planning, decisions, controls, and performance measurement are coherent, compliant, efficient, and measurable across the entire effort.

All Order deliverables and work products shall be developed, produced, maintained, and submitted in full accordance with the requirements, formats, acceptance criteria, and performance standards established in Section F – Deliverables and Performance and the Performance Matrix.

3.1.1 Order Management

The Contractor shall provide order management activities to support the establishment and execution of order governance, management processes, administrative controls, and operations necessary to manage and coordinate all order-administrative tasks as described below.

3.1.1.1 Order Governance

The Contractor shall support establishment and operation of the organizational structure, management processes, controls, and documentation practices required to oversee order execution and ensure compliance with applicable standards, policies, and Government procedures.

The Contractor duties shall include, but are not limited to:

1. Providing the overall management framework, governance, structure, and controls to effectively manage the requirements of this order.
2. Providing a Contractor Program Management Office (CPMO) dedicated exclusively to this order, including a Program Manager and a designated back-up, responsible for executing all management aspects; providing centralized oversight and communication; and ensuring contractor compliance with the requirements.
3. Developing, documenting, implementing, maintaining, and adhering to a Contract Management Plan (CMP).
4. Managing all technical and administrative requirements in alignment with recognized project management standards, including the latest publication of the Project Management Institute's (PMI) *The Guide to the Project Management Body of Knowledge* (PMBOK®), and implementing processes that incorporate governance and decision-making protocols; schedule, cost, and quality controls; issue and risk management; deliverable tracking; change management; and compliance verification.
5. Obtaining formal CO approval before implementing any change to scope, cost, or schedule, and maintaining a detailed Change Log.

3.1.1.2 Staffing and Subcontractor Management

The Contractor shall support the recruiting, onboarding, supervision, retention and oversight of all personnel and subcontractor resources required for performance under this order.

The Contractor duties shall include, but are not limited to:

1. Developing, delivering, and maintaining a Staffing Plan and Subcontractor Management Plan that is to be submitted as a part of the CMP.
2. Developing, delivering, and maintaining a comprehensive Staffing Matrix that is to be submitted as a part of the Monthly Contract Administration Status Report (MCASR).
3. Maintaining an up-to-date Internal Use Software (IUS) report that is to be submitted as a part of the MCASR. The identification of work subject to IUS reporting may change over the lifecycle of the contract and will be coordinated between the CPMO and Government.
4. Recruiting, vetting, assigning, managing, and supervising all personnel working under this order.
5. Providing only **U.S. Citizens** who are capable of obtaining a favorable USCB suitability assessment.
6. Establishing and executing onboarding processes for all personnel, including required documentation, suitability submissions, and access requirements.
7. Initiating and monitoring the Census Investigative Services (CIS) suitability assessment, including conducting quality assurance reviews of all suitability documentation prior to submission to CIS to ensure accuracy, completeness, and timeliness.
8. Obtaining COR notification of a favorable suitability assessment for each staff member prior to commencing work, accessing Government facilities or systems, or charging labor to the order.

9. Providing notification, in writing, to the Contracting Officer's Representative (COR) of personnel being offboarded as soon as possible but no later than twenty-four (24) hours after the exit.
10. Retaining personnel-related documentation and providing such records to the Government upon request.
11. Maintaining the required level of expertise and competency of all personnel by ensuring individuals remain current with evolving technologies and by assuming full responsibility for role-appropriate training and requiring personnel to maintain required professional certifications, accreditations, and proficiencies, at no additional cost to the Government.
12. Providing oversight and tracking for all mandatory USCB and GEO Program training, ensuring all personnel complete required training prior to deadlines specified by the COR.
13. Providing the ability to meet surge requirements by rapidly augmenting staff and responding to unforeseen, ad hoc, or shifting priorities.
14. Assisting the Government with the timely tracking and receipt of Government Furnished Equipment/Government Furnished Information (GFE/GFI), Government system accesses, passwords, etc.
15. Complying with USCB physical and personnel security guidelines.

3.1.1.3 Order Risk Management

The Contractor shall support the proactive identification, assessment, mitigation, monitoring, and escalation of risks and issues that could impact contractual performance, cost, schedule, quality, or compliance. Order risks reported under this subsection shall be included in the Monthly Issue Risk & Issue Report.

The Contractor duties shall include, but are not limited to:

1. Maintaining an active, documented risk management process, including identification, assessment, prioritization, mitigation, and monitoring of all order-execution risks.
2. Developing and maintaining a Risk Register.
3. Developing and executing risk mitigation and/or avoidance plans in coordination with the COR and/or Contracting Officer (CO).
4. Reporting risks, issues, and mitigation plans in all required weekly and monthly reports.
5. Providing immediate notification to the COR and/or CO of any urgent or high-impact risks affecting cost, schedule, performance, compliance, or continuity.

3.1.1.4 Communication Management

The Contractor shall support the establishment of clear, timely, and collaborative communication channels, coordination mechanisms, and meeting cadences between the Contractor and Government to support effective order administration, decision-making, coordination, and documentation.

The Contractor duties shall include, but are not limited to:

1. Establishing and maintaining a Communication Plan.

2. Providing timely notifications of all scheduled and unscheduled events or issues impacting performance.
3. Establishing escalation and crisis-handling procedures for root cause analysis and resolution, to the Government's satisfaction, of major issues.
4. Utilizing Government-approved communication tools and file formats for all reporting and information exchange.
5. Maintaining a Communications and Decision Log to track key communications and decisions.

3.1.2 Program Management

The Contractor shall provide program management activities to support integrated planning, monitoring, control, and coordination of program execution across schedule, quality, risk, readiness, and cross-team collaboration as described below:

3.1.2.1 Program Governance

The Contractor shall support the Contractor's use of Government-issued program governance artifacts, standards, and processes to ensure that all program execution activities conform to GEO requirements, operational frameworks and oversight expectations.

The Contractor duties shall include, but are not limited to:

1. Leveraging and aligning with USCB and GEO governing artifacts, including the: 2030 Census Integration & Implementation Plan (IIP) (Attachment J-XX); 2030 Census Operational Delivery Management Plan (ODMP) (Attachment J-XX); the 2030 Census Test & Evaluation Master Plan (TEMP) (Attachment J-XX); 2030 Census IT Defect Management Plan (Attachment J-XX); GEO Systems Engineering Management Plan (SEMP) (Attachment J-XX); 2030 Census Program Requirements Management Plan (REMP) (Attachment J-XX); and all other applicable GEO program documents, plans, standards and governance requirements.
2. Implementing and executing program change management processes.
3. Storing all deliverables, work products, artifacts, and supporting documentation in the Government-specified document repository accessible to both the CPMO and the Government Program Management Office (GPMO).

3.1.2.2 Schedule Management

The Contractor shall support the development, integration, and maintenance of schedule artifacts and planning inputs required to coordinate work across the order, including the alignment of the Contractor Work Breakdown Structure (CWBS), GEO Work Breakdown Structure (GWBS) (Attachment J-XX), and the GEO Integrated Master Schedule (IMS) (Attachment J-XX).

3.1.2.2.1 Contractor Work Breakdown Structure (CWBS)

The Contractor duties shall include, but are not limited to:

1. Developing, delivering, and maintaining a detailed CWBS.

Tracking, managing, and reporting all contractual deadlines, deliverable due dates, and administrative schedule commitments.

2. Providing and maintaining a CWBS Dictionary.

3.1.2.2.2 GEO Integrated Master Schedule (IMS) Inputs and Schedule Integration

The Contractor duties shall include, but are not limited to:

1. Providing all required schedule inputs to the Government for incorporation into the GEO IMS, ensuring completeness, accuracy, and traceability to the CWBS and associated deliverables.
2. Assigning unique CWBS identifiers to each schedule activity and milestone to maintain one-to-one traceability between the schedule, CWBS, deliverables, risks, and reporting.

3.1.2.2.3 Contractor Schedule

The Contractor duties shall include, but are not limited to:

1. Developing and maintaining an order-level schedule baseline.
2. Reporting schedule variances and schedule risk mitigation strategies.
3. As new development projects are identified, the contractor shall develop a project schedule, an estimate of planned development costs (e.g., Development, Modernization, and Enhancement or DME costs) and track planned vs. Actual costs over the lifecycle of the project.

3.1.2.3 Program Quality Management & QASP Performance

The Contractor shall support the monitoring, control, and continuous improvement of program execution and compliance with Government quality surveillance requirements, including development and maintenance of the Quality Control Plan (QCP), participation in Quality Assurance Surveillance Plan (QASP) (Attachment J-XX) activities and program-level defect prevention and resolution.

The Contractor duties shall include, but are not limited to:

1. Developing, delivering, maintaining, and adhering to a QCP.
2. Assessing and documenting quality assurance/quality control lessons learned throughout the lifecycle of the order.
3. Identifying and reporting risks or issues that may impact QASP performance metrics, including staffing risks, workload changes, system constraints, and process challenges, along with mitigation strategies.
4. Providing advanced notice of any anticipated performance deviations from QASP standards, including justifications, impact assessment, and proposed mitigations.
5. Participating in a Monthly QASP Performance Meeting, as well as any other ad-hoc reviews that may arise during the monthly cycle, to review the Government's assessment of the Contractor's performance.
6. Preventing rework and ability to extend existing functionalities and features by planning and prioritizing appropriately.

7. Ensuring key work products, deliverables and ad-hoc tasks are completed on time and in accordance with established requirements, schedules, and quality standards.
8. Working with government stakeholders and GPMO to track and mitigate urgent defects as they occur.
9. Ensuring that all defects are managed and tracked in a timely manner, that is accepted by the stakeholders and GPMO.

3.1.2.4 Program Risk & Issue Management

The Contractor shall support the identification, documentation, communication, and mitigation of risks and issues that may affect program-level performance or outcomes. Program-level risks reported under this subsection shall be included in the Monthly Issue Risk & Issue Report.

The Contractor duties shall include, but are not limited to:

1. Identifying, documenting, and communicating to the GPMO project risks, mitigations, contingencies, and issues.
2. Supporting/participating in and responding to USCB risk management activities. This includes responding to information requests, supporting processes or procedures, maintaining and/or updating documentation, and attending both scheduled and ad-hoc meetings.
3. Capturing meeting notes and action items for program-level risk meetings that are jointly attended by federal and contractor staff and storing them in a centralized location that is accessible to all appropriate government and contractor management personnel.

3.1.2.5 Data Call and Audit Support

The Contractor shall facilitate the collection, analysis, and reporting of relevant data and information required for order audits or data calls.

The Contractor duties shall include, but are not limited to:

1. Supporting internal and external order audits on work performed under this order.
2. Supporting internal and external (e.g., Department of Commerce (DOC) & Government Accountability Office (GAO) inquiries) data calls related to work performed under this order.
3. Supporting the timely production and presentation of materials in support of GEO data calls and oversight requests (e.g., GAO, Office of the Inspector General (OIG)).

3.1.2.6 Operational Readiness Reviews

The Contractor shall support Operational Readiness Reviews (ORR), which provides the Government confirmation that required capabilities, processes, and resources are prepared for upcoming operational phases or deployments.

The Contractor duties shall include, but not limited to:

1. Scheduling and conducting an ORR with Government stakeholders to present readiness evidence and obtain phase-gate approval.
2. Participating in business and technical interchange meetings and supporting data to ensure agenda items are fully covered.

3.1.2.7 Lessons Learned Management

The Contractor shall support the collection, analysis, documentation, and application of insights to improve processes, coordination, and performance throughout the life of the order.

The Contractor duties shall include, but are not limited to:

1. Executing a Lessons Learned Management process that captures insights, findings, and improvement opportunities derived from all order-administrative activities (e.g., onboarding, reporting, compliance, invoicing.).
2. Capturing lessons learned throughout order execution.
3. Conducting periodic reviews, including at major program checkpoints, to analyze what worked well, what did not, and what changes are required to improve performance, processes, and coordination across Contractor and Government teams.
4. Incorporating lessons learned into updates of management artifacts, including the CMP, QCP, Staffing Plan, Subcontractor Management Plan, Communications Plan, Transition Plans, and relevant execution-level plans.
5. Maintaining a Lessons Learned Log that is version-controlled, and accessible to the Government.
6. Reporting lessons learned to the Government during regular meetings (weekly, monthly, quarterly), or as otherwise directed.
7. Supporting Government-led lessons learned activities, workshops, or after-action reviews by providing data, documentation, and personnel participation.

3.1.3 Transition Management

The Contractor shall support a smooth and efficient onboarding process at the start of the order and an orderly transition at its conclusion.

3.1.3.1 Transition-In

The Contractor shall support a smooth and efficient start of performance by establishing the personnel, processes, documentation, access, and resources necessary to initiate work under this order and to achieve full operational capability without disruption.

The Contractor duties shall include, but are not limited to:

1. Establishing, documenting, maintaining, and submitting a Transition-In Plan (TIP) for review and acceptance.
2. Executing the TIP, with continual tracking of transition activities, including but not limited to status of staff onboarding activities.
3. Participating in the transfer of all applicable knowledge, documentation, artifacts, and operational context from both the Government and, when applicable, from the outgoing Contractor, ensuring continuity of operations and retention of institutional knowledge.
4. Conducting ongoing Transition-In meetings with COR to report progress, address issues, and mitigate risks. Meeting frequency shall be coordinated with the COR based on transition needs.

3.1.3.2 Transition-Out

The Contractor shall support the orderly completion of work at the end of this order by ensuring the transfer of knowledge, data, artifacts, documentation, and responsibilities back to the Government, and when applicable, to an incoming Contractor, while maintaining continuity of operations, safeguarding Government information and assets, and closing out all order-related activities.

The Contractor duties shall include, but are not limited to:

1. Developing, documenting, implementing, maintaining, and adhering to a Transition Out Plan (TOP), that ensures an orderly conclusion to the order and supports transfer of responsibilities to the Government and, when applicable, the incoming Contractor.
2. Coordinating and working in good faith with the Government and/or the incoming Contractor for an orderly and efficient transition of activities performed under this order.
3. Transferring all applicable knowledge, documentation, artifacts, and operational context to the Government and/or the incoming Contractor, ensuring continuity of operations and retention of institutional knowledge.
4. Providing all required close-out documentation, administrative records, and configuration items, including final deliverables, reports, and any other materials requested by the Government.
5. Return all Government-Furnished Equipment (GFE) and Government-Furnished Information (GFI) and ensure removal or secure disposition of Government data from Contractor systems.
6. Coordinating and implementing the TOP with the Government and/or incoming Contractor, providing timely and accurate information, data, and support for close-out reviews and audits.
7. Participating in a coordinated close-out effort to address any final Government concerns and confirming completion of all transition and close-out requirements.

3.1.4 Meetings and Reporting

The Contractor shall support consistent, transparent, and actionable communication of order status, risks, costs, schedule, staffing, and performance through structured meeting cadences and standardized reporting.

3.1.4.1 Kick-Off Meeting

The Contractor duties shall include, but are not limited to:

1. Coordinating and conducting a Kick-Off meeting with the Government at a mutually agreed time and location.
2. Providing meeting materials and status reports to allow for Government review and preparation.
3. Providing the Government with kick-off meeting minutes and an action items list from the meeting.

3.1.4.2 Weekly Status Report and Meeting

The Contractor duties shall include, but are not limited to:

1. Submitting a written weekly status report (WSR).
2. Conducting a status meeting to present the WSR.

3.1.4.3 Monthly Cost and Schedule Report and Meeting

The Contractor duties shall include, but are not limited to:

1. Developing, documenting, and maintaining a Monthly Cost and Schedule Report.
2. Conducting a cost and schedule meeting to present the cost and schedule report.

3.1.4.4 Program Management Review (PMR)

The Contractor duties shall include, but are not limited to:

1. Conducting Program Management Reviews (PMR).
2. Ensuring appropriate technical, operational, and management personnel from the Contractor’s team attend PMRs to support discussion, answer questions, and resolve issues.
3. Producing a list of issues, decisions and action items identified during each PMR and submitting the meeting minutes and action item tracking list to the GPMO.

3.1.4.5 Monthly Risk and Issue Reporting

The Contractor duties shall include, but are not limited to:

1. Developing, documenting, and maintaining a Monthly Risk and Issue Report.

3.2 Task Area 2 – IT Project and Program Management

The Contractor shall provide IT Project and Program Management services that focus on the planning, tracking, management, and deployment of GEO IT programs, projects, and initiatives including but not limited to program planning, governance, government staff training, organizational change management, and product delivery management.

GEO and the contractor must work together to integrate effective program and project management efforts to ensure that performance, schedule, and cost objectives are met. GEO requires expert database consultation and advice on business management integration and project management services that will support and enhance program and project development and management of Geography support services.

3.2.1 Integrated Program and Project Management Support

The contractor shall centrally monitor and manage all services provided under the Order to ensure overall health of the contract and compliance, Performance Measures, and Service Level Agreements (SLAs). The contractor shall provide overarching program and project management of all tasks through the issuance of this Order. The contractor shall perform integrated program and project management in accordance with the Contractor's proposed Management Plan.

The contractor shall:

1. Submit a weekly report of status of MAF/TIGER System database security posture.
2. Provide information on current and recently remediated security findings, Plan of Action & Milestones (POAM) or audit data calls.
3. Report on any escalation required from the federal staff to remove delays or roadblocks for addressing any security issues/findings under issued Orders.
4. Communicate with end-users to gain understanding of requirements.
5. Create software/configuration management documentation.
6. Integrate with formal Change Control Boards.
7. Create and organize storage of supporting documentation (e.g., SharePoint).
8. Create materials to be presented to stakeholders and upper management.
9. Update and maintain schedules in Microsoft Project or in other formats with actual deliverable dates and status for the work assigned.
10. Plan, implement, and report on continuous monitoring activities for security controls of MTDB system components and subcomponents, applications and systems usage to ensure 24/7 support.
11. Utilize typical office automation tools, such as Outlook Web Access (OWA), Microsoft Outlook, Microsoft Word, Microsoft Visio, and Microsoft Excel.

3.2.1.1 Geographic Support Information Specialist and Knowledge

The contractor shall provide technical resources to support the management of the GSP. The SMEs shall have Census expertise and working experience in address frame development, spatial data collection, address data collection, MAF/TIGER System data processing, census geography, MAF/TIGER System data products, and information technology in relation to geographic data collection, processing, and methodologies. Additionally, SMEs shall have expertise in one or more of the following areas: Geographic Programs and Projects, Decennial Census Programs, Geographic-related Organization Support, and Federal and International Organizations.

The contractor shall:

1. Analyze geographic program implementation and provide recommendations.
2. Analyze data and develop recommendations based on the analysis.
3. Prepare technical reports as needed or in accordance with requests by GEO Division Chief, Deputy Division Chief, and Assistant Division Chiefs.
4. Develop technical presentations and supporting materials.
5. Assist in development of Memoranda of Understanding with external organizations.
6. Provide support in development of geoprocessing workflows.
7. Aid with the identification and documentation of technical requirements for tracking, processing, and reporting systems.
8. On a limited and as needed basis, support the GEO Chief in an official travel capacity.
9. Work with the project's stakeholders and subject matter experts to identify and prioritize system requirements.

3.2.2 Government Staff Training

The Contractor shall provide Government Staff Training services to facilitate a smooth transition and maximize the value and impact of the implemented technologies for the USCB.

The Contractor duties shall include, but are not limited to:

10. Developing, documenting, implementing, and maintaining a skills assessment of Government staff.
11. Identifying gaps between organizational skills needed compared to the current skills assessment.
12. Performing a current state assessment of training needs to meet GEO Programs' transformation and application modernization needs.
13. Developing, documenting, maintaining, and updating training plans for Government staff.
14. Developing and delivering training curriculum.
15. Executing and tracking staff training.
16. Assessing the effectiveness of the training plan.
17. Improving training through continuous feedback.

3.2.3 Organizational Change Management

The Contractor shall provide Organizational Change Management services to effectively plan, implement, and manage changes within USCB and to ensure successful adoption and transition.

The Contractor duties shall include, but are not limited to:

1. Developing, documenting, updating, and executing a knowledge transfer strategy and organizational change management plans for federal resources.
2. Providing facilitated workshops to support GEO applicable enterprise and technology transformation initiatives.
3. Developing, documenting, and implementing a training and organizational change management repository.
4. Developing and implementing a communication plan to support technology transformation and application modernization for GEO.
5. Designing, engineering, developing, and executing an integrated approach to knowledge management / information management.
6. Collecting and reporting periodic lessons learned, including organizational change management metrics.

3.2.4 Product Delivery Management

The Contractor shall provide Product Delivery Management services to support the planning, tracking, logistics, and deployment of systems and processes.

3.2.4.1 Operational Delivery Management

The Contractor duties shall include, but are not limited to:

1. Providing IT and non-IT delivery management services including but not limited to planning, tracking, logistics, management, and deploying of various components, systems, and processes.
2. Updating and implementing changes to the Operational Delivery Management Plan (ODMP).
3. Establishing Operation Delivery (OD) teams for each operational delivery.
4. Facilitating OD meetings with an agreed to agenda.
5. Capturing and disseminating meeting minutes and action items related to integration and implementation of operational delivery.
6. Monitoring and managing integration points, interdependencies, and milestones among stakeholders and service / solution providers for product delivery and collaborate to resolve blockers.
7. Investigating and tracking integration and implementation of action items to resolution.
8. Developing and providing executive level briefings and presentation materials for OD, highlighting systems at risk for meeting key Integration and Implementation Framework and Plan (IIP) milestones dates.

9. Monitoring and managing integration points, milestones, and interdependencies among the Decennial Census Program SoS service providers (e.g., contracts, enterprise program).
10. Identifying, documenting, mitigating, and managing OD integration risks and escalating, as necessary.
11. Supporting operational delivery starting from the IBR through closeout and lessons learned.
12. Conducting periodic lessons learned.
13. Collecting and report OD metrics.
14. Conducting phase gate reviews and developing detailed runbooks for software and feature releases into production.

3.2.4.2 Readiness Reviews (RRs)

The Contractor duties shall include, but are not limited to:

1. Supporting and facilitating applicable RR and reporting.
2. Developing, documenting, and maintaining RR materials.
3. Capturing and tracking actions (ensuring all actions have an assignee, have a due date, and have been completed) and escalating and resolving all actions from the RR.
4. Producing summary reports with metrics.
5. Collecting RRs entrance and exit criteria for all systems as identified in the Decennial Census Program Master Entrance / Exit Criteria List prior to each RR.
6. Reviewing and validating all RR entrance and exit criteria as identified in the Decennial Census Program Master Entrance/Exit Criteria List prior to the RR per OD and provide recommendations.

3.2.4.3 Runbooks

The Contractor duties shall include, but are not limited to:

1. Establishing, documenting, developing, and maintaining detailed Runbooks. The activities associated with runbook support include but are not limited to:
   a) Selecting users to participate and coordinate with stakeholders and solution providers.
   b) Monitoring system performance.
   c) Providing a venue for feedback based on predefined success criteria.
2. Facilitating Runbook walkthroughs with service providers and stakeholders.
3. Monitoring and reporting the status of execution of Runbook activities.

3.2.5 Geography Support Services

The Contractor shall provide support for the MAF/TIGER System to include spatial and address data processing and workflow control and management. The DITD provides GEO with support for developing, managing, and controlling interactive and batch processing operations that interact with the MAF/TIGER System.

The Contractor shall:

1. Support and maintenance of the USCB's MAF/TIGER System.
2. Development, Modernization, and Enhancements to the MAF/TIGER System to allow for greater efficiency of automated processing, including the use of artificial intelligence within Census Bureau regulations.
3. Maintenance of MAF/TIGER System applications that exchange data with external partners.
4. Enhancements to improve data quality of the MTDB.
5. Implementation of more cost-effective processes for updating the MTDB and continuing to keep the MAF/TIGER System current.
6. A full range of Information Technology (IT) services in support of the MAF/TIGER and other Geographic Support Program (GSP) initiatives.
7. Research, innovation and adoption of the newest technologies/trends in the industry, including artificial intelligence within Census Bureau regulations.
8. Design and execution of MAF/TIGER application migration to Cloud Service Providers.
9. Develop, baseline and maintain required systems development lifecycle documentation.

3.2.5.1 IT Management and Support

The Contractor shall provide support to GEO and DITD for the implementation of all aspects of MAF/TIGER Systems, such as research, planning, training, applications, documentation, coordination with federal staff, and database administration support. This includes geospatial systems, open-source software, and other Commercial Off the Shelf (COTS) software solutions that make up the MAF/TIGER System. A sampling of the products and services currently used by USCB is included as part of Current Environment (Attachment J-XX):

The Contractor shall:

1. Support databases across four environments, including Development, Test, Stage, and Production. The Government does use enterprise backup services.
2. Provide support in accordance with PWS Section 9.0 – Hours of Operations. This includes support during core hours, after-hours/on call support, regular maintenance support, emergency operations, benchmark support, maintenance, and testing support as required.
3. Resolve issues in accordance with Service Level Requirements (Attachment J-XX_.
4. Be responsible for managing spatial and topology databases in standalone, cluster and high availability modes on on-premises and cloud, to include supporting Exadata environments on-premises as well as Oracle Cloud Infrastructure (OCI), Oracle19c and beyond along with subsequent releases, Oracle Spatial, and Oracle Topology Data Model to support the MAF/TIGER System.

5. Design, develop, and implement complex database processes with Structured Query Language (SQL) and Procedural Language for Structured Query Language (PL/SQL) procedures, functions, and packages in support of Extract-Transform-Load (ETL) functionality.
6. Perform database software installations and upgrades, include patching and upgrading the database software as per USCB mandates and stakeholder schedules. Most work can be conducted during normal business hours, but some maintenance, emergency support, and most patches and upgrades will occur outside of normal hours.
7. Provide maintenance and support on databases hosted both on-prem and in the cloud to include Oracle, PostgreSQL, Aurora, and Amazon RedShift databases. A representative inventory of database instances that are currently supported is included in the Current Environment (Attachment J-XX).
8. Provide geodatabase management services that include knowledge of Oracle spatial and topology, and Oracle on premise as well as in the Cloud, to create, maintain and tune any database within the scope of this contract following industry standards/best practices. These services include identifying the problem in the processes, providing solutions to the problems, providing needed documentation, tuning the database instances and other activities.
9. Support efforts to transition systems from USCB data center to multi-cloud platforms (e.g., AWS GovCloud, Oracle OCI Government Cloud, Google Cloud Platform). This includes:
    a. Performing proofs of concept and analyses of alternatives and presenting recommendations to the Government
    b. Identifying candidate systems, performing system design, implementation and administration of COTS and open-source, and integration with DevSecOps pipeline for the deployment of services, data, and software updates. It includes documentation of the above activities as requested by the Government.
    c. Providing any requested reporting regarding database administration by the requested due date.
    d. Providing fast provisioning of database/instance that adhere to the approved security baseline. and providing/keeping current documentation about each supported application.
10. Ensure all COTS software follows vendor and procurement license agreements.
11. Ensure all systems are tuned, highly available, and perform to meet the GEO product schedules and deadlines.
12. Ensure all systems are monitored using various COTS tools such as Oracle Enterprise Manager (OEM), Esri System Log Parser, SolarWinds, AppDynamics, Sentinel, and MS Defender.
13. Systems are regularly patched to meet the USCB's security mandates and industry best practices.

14. Research and implement the open-source technologies where it is practically possible as part of the MAF/TIGER modernization efforts and based on priorities defined by the USCB.
15. Develop logical and physical architectures.
16. Conduct proofs of concept, prototypes, pilots, analyzes of alternatives in areas of IT pertinent to the vision and requirements of the MAF/TIGER System and where those areas are agreed to be pursued by the Government and the contractor.
17. Perform IT planning activities such as but not limited to capacity planning, scheduling, work breakdown structuring, costing modeling,
18. Ensure COTS systems security is being maintained. This includes working with the appropriate USCB governing bodies and Department of Commerce IT security to establish and maintain security baselines for all system types and ensuring all systems adhere to the documented security baseline.
19. Ensure all systems and servers are available to users with a 99% up-time 24/7 and zero storage or disc space issues.
20. Submit a quarterly up-time report showing the availability of each system.
21. Be responsible for administration and management, including:
    a. Backup/Recovery
    b. Disaster Recovery
    c. Installation
    d. Patching
    e. Upgrades
    f. Migration
    g. Security
    h. Hardening
    i. Troubleshooting
    j. Performance monitoring and tuning
    k. Root cause analysis
    l. Spatial and topology data management support
    m. User support
    n. Spatial/Topology data transfers and refreshes
    o. Manage topology, indices and usage
    p. Automation towards near zero downtime
    q. Proper storage allocation and management
    r. High Availability
    s. Database clustering (Oracle RAC, WebLogic Clustering, Apache load balancing)

t. Fault Tolerance

u. Clustering for application servers

v. Building systems in different availability zones in cloud

w. Support services

x. Detailed reports on systems upgrade procedures and steps.

y. Lessons learned and best approach documents.

z. Tracking of issues and solutions.

aa. Status report on software/server installs, and configuration.

bb. Detailed reports on installation and configuration of systems.

cc. Monthly progress report on systems availability, issues and actions taken.

dd. Security vulnerabilities identified and resolved.

ee. Recommendation on growth and space requirements

ff. List, track, resolve and status of suspected bugs.

gg. List and status of Open Service requests with support.

hh. Project management plan.

ii. Maintenance Activities Monthly (or as agreed).

jj. Systems Maintenance and Upgrades.

kk. Systems backup plan.

ll. Evaluating new technologies and solutions as they relate to GSP's mission and Goals.

3.2.5.2 IT Management and Support for Frames Program

The contractor shall provide geodatabase management services for the Frames Program, which will include PL/SQL development to help their researchers optimize code in a PostgreSQL database. The activities operate in both on-prem and the Enterprise Data Lake (EDL) environments in development, test, and production.

The contractor shall:

1. Work with Demographic Frame staff and Application Development and Services Division (ADSD) to troubleshoot issues on the EDL for the Demographic Frame Project.
2. Review code written by Frames Program researchers and make suggestions for optimization.
3. Recommend and implement indexing, partitioning, and database vacuuming to allow the written code to run more effectively.
4. Have a working knowledge with a database procedural language (Oracle PL/SQL, and/or one or more of the Postgres procedural languages (e.g., PL/pgSQL).
5. Listen to recommendations made by other USCB IT professionals and give input to the Frames Program on the recommended tools and design. This may include researching

new cloud technologies that were recommended by others and helping the team understand how this might impact them.

3.2.5.3 IT Management and Support for Spatial and Topology Applications and COTS Solutions

The contractor shall:

1. Provide support for spatial and topology applications and other COTS software solutions, which includes production deployment, support to database servers, performance maintenance, patch management, help desk support and performance tuning.
2. Provide support for COTS software acquisition and manage license maintenance agreements, migrations, and upgrades.
3. Provide management and administration services to reporting and business intelligence tools.
4. Manage and support application servers for Esri infrastructure.
5. Support users and application developers in SQL, PL/SQL tuning, and troubleshooting.
6. Develop scripts to automate the deployment of application code and facilitate application monitoring.
7. Perform the installation, configuration, patch management, troubleshooting, performance tuning and administration of geospatial and other COTS software solutions on Oracle enterprise Linux, Red Hat Enterprise Linux (RHEL) and Windows Server platforms.
8. Support the installation/integration of geospatial desktop software into the Census Virtual Desktop Environment (VDI).
9. Support continuous monitoring of designated open source and COTS tools.
10. Configure and support golden gate replication of MAF/TIGER for every benchmark operation.
11. Maintain consistent installation and configuration across all environments (development, test, stage, and production).
12. Migrate and deploy applications into various environments.
13. Plan, evaluate and upgrade all MAF/TIGER oracle databases such as 19c to future Oracle releases
14. Provide documented plan each time the migration of Oracle version is required (such as 19c to future Oracle releases)
15. Provision, configure and migrate on-prem databases to cloud.
16. Develop secure baselines for newly adapted open-source databases such as PostgreSQL, aurora, redshift.
17. Lead or assist in data migration from on-prem databases to cloud.
18. Understand terraform code structure and be able to modify configuration files such as "main.cf".

19. Write, plan, and apply terraform scripts to create, manage, modify, and display AWS resources such as RDS, security groups, etc.
20. Ability to use GitLab to promote code and apply in terraform.
21. Submit and manage pull requests from Terraform command line interface.

3.2.6 Governance

The contractor shall:

1. Obtain all required approvals from Census IT Governance including SWG, etc.
2. Comply with Section 508 of the Rehabilitation Act, including the creation and maintenance of wireless portal(s) with content and links.
3. Have FedRAMP authorization or an Authorization to Operate (ATO) letter issued by a Federal Government agency as evidence that they have been assessed and authorized at the Moderate Level according to FIPS 199 and FIPS 200. Post contract award, the Government will perform a review for a local ATO decision, required for go-live, and ongoing security assessments for continuous monitoring thereafter. The Government will determine the risk rating of identified vulnerabilities.
4. Respond to the Government with an action plan and schedule to mitigate security risks found within 10 business days of notification of security risks. The Government will notify the Contractor of whether the action plan and schedule are acceptable.
5. Implement the action plan according to the agreed-upon schedule.
6. Provide a solution that is compliant and operationally capable to support Internet Protocol Version 6 (IPv6), as well as IPv4. Internet Protocol Version 6 (IPv6). Any information technology product or system procured as a result of this Order must be IPv6 Compliant. A compliant product or system must be able to receive process and transmit or forward IPv6 packets and be able to interoperate with other systems and protocols in both IPv4 and IPv6 modes.
7. Provide the ability to restrict government access to the cloud service from government specified networks, in accordance with the Office of Management and Budget (OMB) Memorandum M-08-05, “Implementation of Trusted Internet Connections (TIC),” and the Department of Homeland Security’s “Trusted Internet Connections (TIC) Reference Architecture Document Version 2.0.” If not currently able to provide this ability, the Contractor shall provide the Government with a statement regarding its plans to have such capability, including timeframe.
8. Provide the qualifications and contact information of its security and incident response team.
9. Provide the ability to support user and device 802.1X authentication to authenticate end computers and users before allowing access to the trusted network. Headless devices, such as printers, may be allowed access based on their MAC address.
10. Meet all Federal and department security and posture assessment requirements for wired and wireless users. Posture assessment shall be used to verify that computers meet the DOC

11. Define pre-installation activities required to be completed by the USCB Computer Service Division (CSvD) team to ensure that Development (Dev) and Test platforms are prepared for software installation and configuration.
12. Discuss and receive approval from Census to identify 3rd party product dependencies outside of the licensed products including Hadoop and/or databases. This information is to be communicated to CSvD and other USCB Divisions.
13. Validate with the Census any pre-installation activities identified in a have been completed successfully. These would include, setting up of user IDs, user groups, installing of third-party software, provision of access for the Contractor’s consultants to third party software interfaces and opening of ports for install.
14. Install and configure any licensed software.
15. Run the Contractor’s standard test suite to ensure that any products are installed, configured, and functioning properly.
16. Perform informal knowledge transfer regarding configuration and basic solution sample tasks
17. Provide software development services to support the existing applications, upgrade and/or develop new applications and provide technical support required to support the GSP systems and all upgrades and modernizations efforts on this Order.
18. Provide software maintenance support including all activities which encompass all phases of the SDLC to implement minor enhancements, fix software defects, develop new software, and make changes to take advantage of modern technologies.
19. Adhere to the Enterprise Systems Development Life Cycle when providing software development services (i.e., development, design, testing, etc.).
20. Adhere to all Census IT Governance Requirements.
21. Technical security requirements such as having critical Microsoft patches installed before allowing access to the trusted network.
22. All software upgrades and patching shall be coordinated with the Department’s or individual bureau’s IT department and shall be compliant with each DOC bureaus’ IT security patch policies.
23. Provide the ability to restrict access based on group and provide authorization based on computer AD group membership or other equivalent criteria.
24. EPEAT requirements shall be met as applicable.

3.3 Task Area 3 – Application Development and Management Services

The Contractor shall provide the full spectrum of services for application design and development, architecture, testing, end-to-end IV&V, security, operations and maintenance (O&M), cloud application migration, research and development (R&D), detailed documentation, and innovation. Task Area activities support the development, modernization, and management of applications using cloud native best practices to provide the highest levels of security, quality, and speed.

3.3.1 Application Design and Development

The Contractor shall provide Application Design and Development services to support the design and development of applications that align with GEO's objectives and leverage modern technologies and best practices.

The Contractor shall:

1. Provide the full range of software engineering services to enhance, maintain, and/or refactor existing applications and design and develop new applications.
2. Provide software engineering services using USCB and Commerce Department software development lifecycle frameworks and guides, as well as industry best practices and methodologies such as, but not limited to Agile, DevOps and DevSecOps.
3. Document and formalize industry best practices to be incorporated and/or adopted as USCB methodologies.
4. Provide modern software engineering approaches, including the use of Artificial Intelligence that meet federal regulations, that enhance speed, security, and value of applications and service delivery including but not limited to approaches for emerging trends and technologies such as platform engineering, infrastructure as code/everything as code, and the use of generative AI for code development and/or testing.
5. Incorporate functional and non-functional requirements in designs.

3.3.2 Application Architecture Support

The Contractor shall provide Application Architecture services to support the design and definition of the overall structure and framework of the applications.

The Contractor shall:

1. Provide solution/application-level architecture support in coordination with the USCB's enterprise architecture.
2. Develop, document, and maintain design patterns including but not limited to service levels, service-oriented architecture, and application programming interfaces (APIs).
3. Develop and maintain Concepts of Operations based on business requirements by application.
4. Identify, document, and assist the government in the development of microservice specifications aligned to overall USCB vision for service-oriented architecture.
5. Provide best practices and solutions that meet USCB functional and non-functional requirements to help migrate the USCB application portfolio into a cloud-based, service/mesh-oriented architecture, comprised of micro/mini services and coarse/granular APIs.
6. Assist in troubleshooting or identifying solutions to reduce costs and/or improve performance for compute, compute functions, high performance computing, storage, or other services.
7. Assess program needs to determine whether new applications need to be achieved through GOTS, COTS or other application types.

3.3.3 Application Testing

The Contractor shall provide Application Testing to ensure the quality, reliability, and functionality of the applications and software being developed or modernized inclusive of integration application testing, in support of quality control and quality assurance of applications.

The Contractor shall:

1. Provide application-level software testing including but not limited to performance testing, scalability testing, integration testing, exception testing, pairwise and security testing.
2. Develop and perform automated testing services.
3. Develop an end-to-end test management approach that supports requirements traceability, defect analysis, and retrospectives throughout the software delivery pipeline.
4. Develop, update, maintain, and disseminate Test Analysis Reports.
5. Collaborate with application developers on defect resolution including but not limited to categorizing, tracking, and dispositioning all software defects.
6. Provide application and database test execution support through manual and automated testing.
7. Create and maintain business threads/steps from workflows to support the application testing activities.
8. Develop, update, maintain, and disseminate a Defect Management Plan.
9. Develop test plans including but not limited to application testing, automation, end to end testing, integration testing, data validation, system integration, output testing, and exception testing.
10. Create test data to support application testing.
11. Conduct multi-layer testing (including the validation of outputs) on multiple platforms (e.g., mobile devices on iOS and Android, laptops, tablets, and desktops).
12. Conduct application testing in the integrated environment across all relevant hosted environments.
13. Support and participate in reviews such as test readiness and test analysis reviews.
14. Develop and maintain test cases, test scripts, test scenarios, and test metrics.
15. Maintain source code and documentation repositories.
16. Facilitate the release of software from the development environment to testing, staging, user acceptance testing, and production environments.

3.3.4 Application Security

The Contractor shall provide Application Security services to ensure the protection and integrity of applications against potential threats and vulnerabilities.

The Contractor shall:

1. Provide application development and management services in accordance with all federal IT security guidance, policies, and regulations applicable to the Commerce

Department and the USCB including but not limited to Zero Trust and the USCB Risk Management Framework.

2. Develop and implement specialized security solutions, toolsets, toolkits, frameworks, threat modeling assessments, vulnerability assessments, guides, and processes.
3. Support development teams with security testing, to include test plan development, security testing services, and documentation/reporting for USCB applications.
4. Develop, document, implement, and maintain an application security program, identify stakeholders, define security requirements, and establish metrics and reporting to track progress and improvement over time.
5. Support the development and management of security related documentation, including, but not limited to, Interagency Security Agreements, Software Bill of Materials, standards for systems, application, and data, and playbooks related to secure coding, security integration, security operations, and incident response.
6. Scan application components to detect outdated and/or vulnerable libraries and license issues in third-party libraries.
7. Train and educate developers and testers on security best practices and identifying common security vulnerabilities found in software and applications.
8. Develop and implement best practices to include, but not limited to secure coding practices, secure session management, secure file handling practices, secure communications protocols and encryptions, secure error handing, exception management, logging, and monitoring mechanisms.
9. Respond to security-related data calls from, including, but not limited to, Office of Information Systems, and Federal Information Security Modernization Act (FISMA).
10. Consult with stakeholders, such as procurement and legal teams, to include appropriate security clauses and requirements in contractual agreements with third-party vendors.
11. Assist in the evaluation and selection of new tools and applications and provide written documentation of assessments, findings, and recommendations.
12. Implement mechanisms for continuous monitoring and assessment of third-party software components to ensure ongoing compliance with security requirements and timely identification of any potential vulnerabilities or risks.
13. Develop and implement secure software distribution and update mechanisms, including secure repositories and digital signatures, to ensure the integrity and authenticity of software components to include supply chain attestation.
14. Support Continuity of Operation Plans (COOP), Information System Contingency Plans and Disaster Recovery (DR) that address software continuity and comply with National Institute of Standards and Technology guidelines and the Bureau's IT Security program.
15. Develop, document, implement, and maintain COOP and DR strategies, architectures, as well as designing and conducting tabletop exercises and simulations to validate the effectiveness of COOP and disaster recovery plans.
16. Provide assessments of these tabletop activities, document recommendations, and perform necessary training for stakeholders based on these recommendations.

3.3.5 Application Operations and Maintenance

The Contractor shall provide Application Operations and Maintenance Execution services to ensure smooth and efficient operation, maintenance, and support of USCB Program software and applications.

The Contractor shall:

1. Handle release management by planning and coordinating releases, ensuring smooth transitions to all environments, documenting processes, monitoring progress, and conducting post-release reviews while managing risks, ensuring compliance with policies.
2. Develop and maintain systems architecture and engineering artifacts, in alignment with USCB Enterprise Architecture.
3. Collaborate with IT service management and enterprise infrastructure teams to monitor application performance.
4. Provide operations and maintenance support of applications in all environments including but not limited to maintenance requests such as patching, defect resolution, minor enhancements, development workflow and providing updated information to the appropriate stakeholders.
5. Collaborate, participate, and be available for rapid response teams, and participate in Root Cause Analysis and troubleshooting to assess and diagnose problems, identify root causes of technical problems, and recommend solutions.
6. Establish and implement procedures, metrics, and documentation on defect management and technical debt reduction.
7. Collaborate with USCB stakeholders to minimize impacts to the mission due to planned disruptions due to system updates including but not limited to change management policies and procedures.
8. Maintain, implement, and/or operate backup and recovery plans in collaboration with system, application, and infrastructure stakeholders.
9. Assist in performing cost analysis including but not limited to total cost of ownership costs, infrastructure and/or cloud costs, and other O&M costs.
10. Create and maintain custom reports, graphics, and dashboards to support application operations and monitoring.
11. Provide hardware infrastructure support and services.
12. Provide administration support for COTS, GOTS, open-source, SaaS and in-house developed software.
13. Conduct reviews, audits, and risk analysis and provide findings and recommendations.
14. Provide operations support for applications including but not limited to Tier 3 support. The Contractor must track support activities and metrics to be used by the government to assess service level quality.

3.3.6 Cloud Application Migration

The Contractor shall provide Cloud Application Migration services to support the planning, architecture, and execution of the movement of applications or workloads from on-premises infrastructure to external cloud services, or between different external cloud services.

The Contractor shall:

1. Support the USCB in articulating a cloud application migration value proposition defined for both business and IT.
2. Establish, implement, and maintain a cloud application migration strategy and roadmap that aligns to USCB IT strategic goals, enterprise architecture roadmap, and security guidelines.
3. Establish, implement, and maintain decision principles for migration, informed by application and team readiness, business priorities, application modernization strategies, and vendor capabilities.
4. Establish, implement, and maintain an application rationalization/optimization plan and migrate applications to the cloud.
5. Develop, document, implement, and maintain a cloud adoption framework.
6. Develop, document, implement, and maintain a cloud application governance process.
7. Establish, implement, and maintain the appropriate cloud footprint optimization and evolution capabilities for ongoing FinOps, DevSecOps, and Zero Trust Model disciplines for cloud application migration.

3.3.7 Research, Development, and Innovation

The Contractor shall provide Research, Development and Innovation services to support activities to assess the technical, operational, cost feasibility, and analyses of alternatives (AoA) of new technologies and approaches to meet functional and non-functional requirements.

The Contractor shall:

1. Perform evaluations and/or analysis of alternatives of new or emerging technology applications for requirements including but not limited to evaluations such as technical feasibility, operational feasibility, cost, scalability, security, and risks.
2. Design, develop, document, and implement application prototypes and pilots to determine feasibility of technologies, emerging, or otherwise. The development and/or configuration of pre-production applications, COTS/GOTS, or open source-based prototypes may be required.
3. Assist in the evaluation of the current environment and make recommendations on a research environment that will allow for the analysis of research and development activities.
4. Perform engineering studies and analyses such as, including but not limited to, demonstrating the feasibility of conceptual approaches, evaluating design risk, evaluating the viability of potential solutions, alternatives to various technical issues and challenges, emerging products, or technology, and comparing and analyzing trade-offs.

5. Perform technology related studies and analyses including but not limited to logistics, supportability, engineering, financial, operational, business processes, and applications to include mobile applications, modernization of existing systems and applications, interoperability, and information sharing.

3.3.8 Geography Support Services

3.3.8.1 Workflow Control, Testing, and Configuration Management

The Contractor shall provide Workflow Control, Testing, and Configuration services to support the design, development, testing, implementation, and maintenance of the system components and functionality that provide the framework for testing, production control, and release management.

3.3.8.1.1 Workflow Control

The Contractor shall provide Workflow Control and Management services to support managing, controlling, and reporting for batch processing operations that interact with the MAF/TIGER System and for large GEO operations that have multiple project phases. This is accomplished using both COTS software as well as custom-developed applications.

The contractor shall:

1. Design, develop, implement, unit test, support, maintain and enhance automated WEB enabled batch control systems in the ONPREM and Cloud environments.
2. Research available open source/COTS applications for implementation of a reimagined workflow system in the Cloud.
3. Design, develop, implement, unit test, support, maintain and enhance automated Production control systems in the ONPREM and cloud environments.
4. Provide plans and documentation for the design and implementation of automated production workflow control and reporting systems.
5. Develop customized web-based user interfaces for each operational specific PCS that will typically have screens to provide for log-in, search, comments, posting of dates, generation of reports, and other operation-specific activities.
6. Maintain and provide enhancements to the existing PCSs in the ONPREM and cloud environments.
7. Maintain and provide enhancements to Web-based Control Systems (WebCS) that manage batch production processing for a variety of production, update, and benchmarking operations.
8. Maintain and provide enhancements to the WEB enabled batch Control System (WEBCS) core engine in the ONPREM environment.
9. Provide operational support of batch geoprocessing jobs that include the design, scheduling, and coordination of processing strategies, reporting of incidents and defects, and the coordination of responses to the batch processing.

10. Create and maintain design documentation following industry standards for each production control system as well as an overall design document for WebCS.
11. Continue work to refactor PCS's to microservices and migrate to the AWS cloud as determined by the Government.
12. Establish and implement a schedule for all geographic operations PCSs for which PCSs will be transferred to an open-source database, and which will be migrating to the cloud with new architecture.

3.3.8.1.2 Independent Testing and Validation

The Contractor shall provide Independent Testing and Validation services to support the GSP with performing testing and validation of internally developed software and systems. This includes evaluation of System Requirement Specifications and User Stories, the development of test cases, the development and maintenance of independent automated testing scripts for incorporation into the CI/CD pipeline, and test reports for the myriad of interactive and batch software that comprise the MAF/TIGER System.

The Contractor shall:

1. Conduct independent testing and validation of internally developed software and systems, and internal implementations of COTS software utilizing established processes and procedures while suggesting and adopting industry best practices as necessary, that includes the preparation of test case designs, execution of test cases, coordination of testing activities and reporting of results.

3.3.8.1.3 Support for Automated Testing

The contractor shall:

1. Evaluate software requirements to develop/update test cases and other test reports for the myriad of interactive and batch software that comprise the MAF/TIGER System according to established standards and guidelines, adopting industry best practices as necessary, for use in independent testing and validation efforts.
2. Create and maintain test cases and automated test scripts (currently using GitLab and Katalon) and conduct independent software testing according to established project schedules, guidelines and industry best practices for each MAF/TIGER System application (Red Hat Enterprise Linux [RHEL] processing [batch] and web-based applications) that will undergo development (new application, redesign, or Cloud transformation) as agreed upon by the U.S. USCB.
3. Develop, update, and maintain custom scripts (yaml, shell, etc.) according to industry best practices for CI/CD pipeline in GitLab to incorporate independent software validation automated testing scripts and report software validation testing results for each MAF/TIGER System application (Red Hat Enterprise Linux [RHEL] processing [batch] and web-based applications) that will undergo development (new application, redesign, or Cloud transformation) according to established documented the corresponding software application project schedule as agreed upon by the U.S. USCB.

4. Coordinate the integration of automated test scripts for independent testing and validation into the CI/CD pipeline with all required teams according to the established documented corresponding software application project schedules as agreed upon by the U.S. USCB.
5. Be a member of the DevSecOps team for which the test cases, automated test scripts, and independent testing are needed. This includes testing in the development environment during Agile Sprints and in the Test Environment when software is released as agreed upon by the U.S. USCB.
6. Ensure any new DevSecOps application shall have independent automated test scripts built and integrated into the CI/CD pipeline within the project's documented schedule.
7. Ensure all existing applications that are integrated into the CI/CD pipeline are leveraging independent automated test scripts that are executed within the CI/CD pipeline to support independent software testing. Where none exist, the contract shall build new automated test scripts and ensure integration into the CI/CD pipeline within the established schedule as agreed upon by the U.S. USCB.
8. Maximize the use of automated testing within the CI/CD pipeline and reduce the dependence on manual testing for independent testing and validation to the furthest degree feasible and according to established and documented software application project schedule.
9. Provide training to government and contractor staff on performing all related interactive and DevSecOps automated testing activities.
10. Ensure the proper security scans are run on the automated testing scripts and shall resolve security vulnerabilities found in the automated testing scripts within established schedule as agreed upon by the U.S. USCB.
11. Research, implement, and manage open source and COTS solutions (example: Katalon, Selenium) assuring alignment with industry best practices for automated interactive and batch DevSecOps testing.
12. Complete evaluation of new automated testing tools for batch processing and interactive applications with associated documentation.
13. Complete implementation of new automated testing tools for batch processing and interactive applications upon agreed, documented schedule when the USCB agrees with recommended tools.

3.3.8.1.4 Change and Configuration Management

The Contractor shall provide Change and Configuration Management services to support designing, implementing, managing, customizing, and enhancing processes and automated systems that are used to manage changes to both the hardware and software environments used by the GSP.

The Contractor shall:

1. Provide document management support that includes: reviewing documents related to the change management process for accuracy and completeness, issuing document identifiers for Software Development Lifecycle (SDLC) documents, maintaining the document management log, performing quality control (QC) checks on documents that impact change, providing guidance to GEO and/or DITD staff on how to resolve issues

discovered during the document QC checks, corresponding with GEO and DITD staff to resolve questions related to documentation, baselining and posting ADC/CCB approved documents to the official electronic library, and ensuring staff are following the established change management policies and procedures for document management.

2. Provide software configuration and release management support services that include: facilitating software release, technical review board and change control board meetings, ensuring NIST controls are followed when releasing products to the non-development environments, performing QC checks on version description/release documentation for internally developed software, providing guidance to GEO and/or DITD staff on how to resolve issues discovered during the QC of version description/release documentation, and managing the established COTS-based systems that are used to track and deploy software to GSP's processing environments.
3. Provide design, development, implementation and ongoing operations, maintenance, and enhancement support for operations/functions in a software configuration and release management system and a COTS-based change management system.

3.3.8.1.5 Data and Process Enhancement and Verification Software

The contractor shall:

1. Provide support for enterprise verification software under the general task areas of source code development, testing (including support of ITV testing), documentation, production support, maintenance, and end-of-life (closeout). Examples of application software projects to support data quality and enterprise verification efforts are provided in the Current Environment (Attachment J-XX).
2. Maintain, develop, and enhance new software as needed to perform verification and validation of geographic products
3. Automated batch quality control software to verify receipt, loading, and aggregation of decennial tabulation population and housing counts.
4. Automated processes to verify correct migration and loading of geographic areas.
5. Stage Summary Level data to support quality control efforts, e.g., of Geodatabase, Cartographic Boundary Files, and SAIPE File Geodatabase products.

3.3.8.1.6 Quality Control Dashboard Application

The Contractor shall provide support for the Quality Control Dashboard application, a centralized reporting dashboard that collects the results from all QC applications

The contractor shall:

1. Maintain the Quality Control Dashboard application. The application shall provide a web interface via which all QC results for all products can be found, reviewed, and managed by subject matter staff.
2. Design the interface framework to integrate QC applications with the QC Dashboard.
3. Assist on the modification of the QC applications to integrate with the QC Dashboard.

3.3.8.2 Address Processing

The Contractor shall provide services to support the development and maintenance of the MAF and address-related software, as well as various customer products and systems that utilize information from the MAF.

The contractor shall:

1. Design, develop, enhance, debug, and implement batch and interactive software to perform address data management.
2. Research, test, build, and coordinate the conversion and/or integration of new modules/API's based on client requirements to add/enhance functionality and quality of address data processing.
3. Consult with project teams and end users to identify application requirements.
4. Perform feasibility analysis on potential future projects to management.
5. Assist in the evaluation and recommendation of application software packages, application integration and testing tools.
6. Resolve problems with software and respond to suggestions for improvements and enhancements.
7. Effective senior-level communication, verbal and written, with the customer's managers and others.
8. Adaptation of Agile practices for all development, release, and deployments.
9. Implementation of process improvements for long running and complicated processes (e.g., geocoding, map spot numbering.)
10. Reengineering architecture and implementation for critical update processes such as DSF Refresh, Geocoding.
11. Adapt and implement secure DEVOPS for all development, release and deployments that includes automated testing.
12. Continuous improvements in new parsing, standardizing and matching and search technologies (Elastic Search) for Address Matching and research purposes.
13. Implement processes to automate inspection of software for code quality, bugs, vulnerabilities, secrets, and best practices in software design and development.
14. Design, develop and implement tallies, informational web pages and/or dashboards describing the status of update programs, product creation, and contents of the MAF etc.
15. Design, develop, implement and maintain DisplayMAF applications used internally by programmers and SMEs in the address area to support our day-to-day operations.
16. Design, develop, implement and maintain automated and manual QC and data fix software for all address update software and address products.
17. Create a catalogue and/or inventory of existing Address benchmarks, Address products and develop a records management policy. Review documentation for all existing Address processing systems/software and create/maintain as necessary.

3.3.8.2.1 Address Update Systems

The Contractor shall provide services to support Address Update Systems. Address Update Systems provide the architectural design, development, implementation, and day-to-day management of the large-scale address database system. This area is responsible for systems design, systems development, test and implementation of address matching, address standardization, geocoding, DSF refresh software/services for updating MAF with data received from USPS, tribal, state, and local governments, and USCB field operations.

The Address Update Software and Systems sub-area provides the support to create, maintain, and update internal address products using the MAF and the data from the U.S. Postal Service's DSF, partner files from various government entities, and data collected from field operations.

The contractor shall:

1. Provide design, development, implementation, maintenance, and enhancement support for software applications, and systems that update the MAF as well as maintain linkages to geocodes and other spatial data.
2. Provide design, development, implementation, maintenance, and enhancement support for new pre-process data verification edits. This includes the internally developed software API, interfaces that facilitate data exchange with other systems, and software that facilitates evaluations of address data.
3. Design, develop, enhance, and maintain solutions that perform address geocoding as address and spatial update occur on a flow basis not at time of product creation.
4. Design, develop, enhance, and maintain solutions that perform address standardization and address matching using new technologies and trends in the industry.
5. Design, develop, enhance, and maintain software to enforce the rules of edits associated with the maintenance of the address component of the MAF/TIGER System not only as a batch processing but also on a near real-time basis system.
6. Develop, enhance, and maintain services that enable ingest of address/structural updates into MTDB from external systems, and/or external data sources.
7. Design, develop, and implement systems to geocode on demand.
8. Participate in improvement of software, service, tool, and research related to MAF Address Parsing, Standardization, Matching, Geocoding, and Update in batch, near real-time and as a service. Examples of these improvements are Address and Spatial Matching for Parcel Data, Enterprise Address MAF Update, and MAF Restructure for BSA Matching.
9. Design, develop, enhance, and maintain software that performs address matching and geocoding to create products for external customers such as SAIPE Geocoding, NHIS MSG Geocoding, and Zip+4 County Coding Guide.
10. Upgrade/migrate software systems to latest versions of Oracle and Java to be compliant with OIS and Vendor support maintenance agreements.
11. Support research and restructure of MAF in support of GSS and Decennial Census that includes Census tests. Provide documentation of the above activities as outlined in the deliverables.

3.3.8.2.2 Batch and Interactive Geocoding and Matching Services

The Contractor shall provide services for geocoding software and systems that performs or assist with geocoding operations.

The contractor shall:

1. Design, develop, implement, and maintain software to assign geographic locations to addresses via automated geocoding.
2. Design, develop, implement, and maintain software to assign geographic locations to addresses via an interactive clerical geocoding.
3. Design, develop, implement, and maintain software to provide automated address matching services.
4. When needed, provide updates to the Parcel Address Matching Sub-team (PAMS) operation
5. Design, develop, implement, and maintain a web-based interface to support address geocoding and matching requests, and web-based transmission of assigned geography and address matching results.
6. Design, develop, implement, and maintain software to support the American Housing Survey geocoding. This includes spatial matching against parcel data, address matching against parcel data, and identifying geographic area codes across multiple time periods.
7. Ensure MAF update software captures information about the updates including proposed creation, modification, or deletion of address records during and after address update.
8. Facilitate research for both internal and external sources to support accurate geocoding.
9. When needed, utilize external batch geocoding services (such as CODE1) to geocode various address files for inclusion in the MAF or for other, outside the MAF, uses.
10. Provide Economic Census Geocoding support to design, develop, implement, and maintain software to support interactive clerical address matching and geocoding services in support of the ECON Census so that it is in accordance with the Geographic Area Reference File (GARF), which lists the nested hierarchical areas ranging from Nationwide to Place-level.
11. Provide documentation of the above activities as requested by the Government.
12. Support in research, design, development and implementation of migrating Header Coding Service, Address Range Geocoding Service and Applicant Geocoding Services to designated Cloud environment AMT in DMZ and/or non-DMZ.
13. Create and maintain system level documentation in GitLab with Architecture, API Documentation, System Requirements and other documentation as assigned by government.
14. Contractor shall develop a plan and perform migration of Address Range Geocoding Service and Header Coding Service to cloud (DMZ, non-DMZ and/or EDL if applicable). Both applications shall be refactored as cloud native, ensuring a cost-efficient design.
15. Improve the header coding process used to validate ZIP from address with city/state/ZIP. This process is used during the geocoding process for better matches.

The existing system utilizes legacy code that can be updated to a coding language for ease of maintenance and logic updates. Plans include:

a. Convert the use of flat files and load data into oracle tables.
b. Rewrite c code into a modern coding language such as Python, java.
c. Research the use of CODE-1 software to understand the value.
d. Update the process flow to increase throughput.

3.3.8.2.3 ACS Automated and Clerical Geocoding

The Geocoding Systems Branch (GSB), with the support of contractors, provides ongoing production level support for regular surveys throughout the decade, most notably for ACS geocoding. This includes occasional work for Survey of Income and Program Participation (SIPP) and Current Population Survey (CPS) surveys, but ACS is by far the largest customer. GSB typically plans enhancements of software and reference files needed for ACS processing on a yearly basis. There is additional work on this post-2030 to use the newly tabulated geography. This work would be conducted starting late in the calendar year to deploy in March of each year. There are several major categories of software development that would provide improvement to the automated portion of ACS geocoding.

The Contractor shall:

1. Full review and analysis of the header coding process. Assess the value added by using CODE-1 and determine the need for 100 percent header coding in block coding activities.
2. Research alternatives for CODE-1 to support modernization effort of migrating to Cloud environment.
3. Full integration and maintenance of interfaces for consistent look and use of Computer Assisted Clerical Coding (CACC) under the MaCS umbrella by updating user interface, to facilitate user training and software maintenance, in support of ACS for Place of Work geocoding and clerical coding jobs.
4. Annual updates of CACC and SUMO to support ACS geocoding.
5. Full review and analysis of employer matching process. Determine feasibility of transferring major processes to database instead of on the batch server.
6. Refresh of master employer file. Through coordination with GEO and SEHSD area, assemble requirements, review sample files, and integrate new files.
7. Document steps for maintaining monthly ACS and update GSB inventory.

3.3.8.2.4 Matching and Coding Software (MaCS)

MaCS is an interactive web interface that encompasses different modules that are built for the needs of specific operations. All modules have one or more of these functionalities: user account management, workload management, clerical data entry, text search of MAF addresses, address ranges, and other references, map interface, reports, and quality control. There are also training datasets that are typically needed.

The contractor shall design, develop, test and implement future enhancements to MaCS that could include:

1. Issues Tracking and Reporting Service (ITaRS)
    a) Development and maintenance of a service to store address and spatial referrals.
2. MAF/TIGER Referral System (MTRS)
    a) Development and maintenance of a stand-alone MaCS module providing functionality to support address and spatial related referrals.
    b) Development and maintenance of a stand-alone MaCS module providing functionality to support address and spatial related referrals from within preexisting applications (JAGUAR, ARC, MAF Browser, other applications).
    c) Development and maintenance of functionality allowing user to enter final resolution results from JAGUAR edits into MTRS and send results back to ITaRS.
    d) Development of functionality in MaCS to create an ADDUP File from address referral records that resulted in changes needed to the MAF.
3. Optimus MaCS/MaCS Builder
    a) Develop a “mega” MaCS service that includes fields from table commonly used across all current and past MaCS operation specific modules.
    b) Development and maintenance of a reusable customizable MaCS module that is based on functionality commonly used across all current and past MaCS operation specific applications.
4. Update projects to use CI/CD process with Gitlab.
5. Create a general address review tool in the cloud environment for MaCS on an agreed upon documented schedule.
6. Utilize MAGE EDL environment and leverage lambda, event bridging and chron jobs in building the application.
7. Communicate efficiently with project owners on the progress.
8. Research efficient ways of implementing communication between MAGE and CDAC. Design and implement in MAGE environment with backend being MAGE database and communicate to CDAC using REST protocols.
9. Communicate architectural changes made and upload to share point in timely manner.
10. Document challenges encountered, architectural overview, system requirements, API documentation and requests submitted for requesting a new EDL environment.
11. Work with external dependencies to meet target dates of Development, ITE, Production and keep team abreast of the communication.
12. Coordinate and be available when there are any research/implementation topics or discussions.

3.3.8.2.5 Census Geocoder

The contractor shall design, develop, test, and implement enhancements to the geocoding services upon which the Census Geocoder is built. This is a web application/REST service that will take an address or batch file of address and geocode the address(es) returning the match status and location/geographic data. Geocoding services also contain a batch component. The batch consists of bash and python scripts that are run by WPSB and perform the operation of splitting a large address file (over 10,000 addresses) into file sizes that the geocoder web service will process, submits them, and recombines the results. Future planned enhancements to the geocoder include:

1. Update Census Locator using the latest release to improve geocoding. New files and the current application will need modifications to use the updated locator service.
2. Update functionality for handling error messages.
3. Update locator service with new benchmark.
4. Update user experience for reporting issues.
5. Improve user interface based on customer feedback.
6. Update the service to use cloud-native solution.
7. Update the service to enhance Puerto Rico searching capabilities.
8. Update the service to support queuing system for customer's requests.
9. Maintain the application and upgrade in timely manner with agency approved version of the software.

3.3.8.2.6 Address Products

This area provides periodic and single-use address products, derived from the MAF, in a variety of formats, to customer systems. A list of standard address products created on a regular basis is provided in the Current Environment (Attachment J-XX).

The contractor shall:

1. Provide design, development, implementation, maintenance, and enhancement support for software systems and databases that are utilized to view and produce products from the address data stored in the MAF.
2. Develop and maintain map spot numbering software to implement spatial numbering of address coordinate locations in the MAF/TIGER System.
3. Design and develop enhancements and maintain software that creates valid ZIP Code to County relationships.
4. Develop and maintain software that assigns Address Characteristics (i.e., city style vs non-city style, supported by the Delivery Sequence File) at the block level.
5. Maintain software that ingests, processes, and stores the City-state File for each benchmark cycle.
6. Implement automated processes to assure the quality and consistency of address products.

7. Design, implement, develop, and maintain services for provisioning address data through service-oriented architecture.
8. Support provision of address products to customers in the USCB's Enterprise Data Lake (EDL), using tools and applications native to the EDL such as Postgres, Amazon Redshift, or Apache Hive.
9. Implement a design for requesting products that meets the USCB data stewardship policies.
10. Provide custom tallies/dashboards of MAF/TIGER statistics, program/project statuses and geographic partnership programs.
11. Create and host standard products in multiple formats such as pipe-delimited, text, and Oracle.
12. Provide documentation of the above activities as requested by the Government, review documentation for all existing Address processing systems/software and create/maintain as necessary.

3.3.8.2.7 Address Management Systems Enhancement

The Contractor shall:

1. Enhance MAF Features as a service – modify the system so LiMA can consume the data instead of delivering.
    a) Create an expansion of Address Provisioning service to consume address data inside an application.
    b) Create the ability to generate a product for a unit of work.
2. Modernize and Maintain MAF Browser
    a) Continue Application Programming Interfaces (API) and other system updates to ensure improved functionality and cloud compatibility of the new MAF Browser system and provide functionality to be included in other applications.
    b) Develop and implement a map viewer used universally by many address and/or spatial systems. This will improve usability by providing enhanced visualization, which is something the MAF Browser has never had. If the user can visualize MAF and spatial data, then analysis becomes much easier and less prone to misinterpretation.
    c) Create modules and interfaces with the address data to improve and maintain the search function and speed of queries.
    d) Migrate MAF Browser tool to the cloud environment on an agreed upon documented schedule.
3. Support annual updates of CACC and SUMO to support ACS geocoding.
4. Event based geocoding: Enhancement of Geocoding system with ability to recompute geocodes whenever boundaries and feature updates are made in our database, so this activity can be moved out of our benchmarking process resulting in reduction of 10 days of lock in period.
    a) Ability to geocode at granular level.

   b) Maintain/Rerun geocodes up to date based on spatial/address change.
   c) Run computation on daily basis similar to DTD.
   d) Geocoding as a service
5. Support upgrade/migration/rearchitect/refactor existing systems from on-prem to Cloud, as requested by GEO and DITD.
6. Provide capability to view across databases and show differentials.
7. Research and implement Address standardizer enhancements.
8. Modernize Address Standardizer:
   a) Review all different address standardizers.
   b) Consolidate functionality into a single solution.
   c) Make this a service.
9. Architect, design, develop, implement, and maintain suite of current and new address-related functionalities as part Address Management Tool (AMT).

3.3.8.3 Geographic Areas Operations, Spatial Data Processing, and Products

The Spatial Data Systems Functional Area provides GSP with support for the core applications and systems that directly access, manipulate, manage, and store the data in the MTDB.

3.3.8.3.1 Spatial Update Applications

The Spatial Update Systems sub-area provides GSP with support for the development and maintenance of automated processes that update the spatial data in the MTDB and other supporting schemas in the MAF/TIGER System.

The contractor shall:

1. Provide design, development, implementation maintenance, and enhancement support for software systems that perform automated updates to the features in the MAF/TIGER System.
2. Provide design, development, implementation, maintenance, and enhancement support for software systems that perform automated delineation of new entity layers in the MTDB.
3. Provide documentation (design, software code, and system) for the above activities.

3.3.8.3.2 Core Update Applications

The Core Update area provides centralized low-level software to support updates to the MAF/TIGER System and keep it consistent. Software is detailed in the Current Environment (Attachment J-XX).

The contractor shall:

1. Maintain all documentation including the following:
   a) CoreAPI

   b) Delayed Calculations Package
   c) Day-to-Day System
   d) MAF/TIGER Utilities (MTUTIL) schema
   e) Map Spot Numbering Software
2. Maintain the MAF/TIGER System CoreAPI libraries to ensure low-level updates to the MTDB maintain MAF/TIGER System's integrity and consistency.
3. Enhance and expand the Business Rules engine, including provision for improved user interface and flexibility for end users to adjust, enable, or disable rules on–the-fly.
4. Maintain and enhance the TIGER Metadata (TMeta) system. TMeta keeps track of the quality of geospatial source data used to update features in MTDB. This information is used to prevent updates using source data of lower quality than that of the feature in the MTDB.
5. Maintain and enhance the Geographic Area Edits, providing additional flexible interfaces for the end users to customize results and outcomes.
6. Improve timeliness of feedback to and flexibility for end users to test, update, and run business rules and geographic edits on demand to decrease required time for edit implementation and resolution.
7. Provide documentation of the above activities.

3.3.8.3.3 Spatial and Geographic Products

The Spatial and Geographic Products sub-area is responsible for derivation of spatial and geographic products from MAF/TIGER, such as geographic reference files, spatial extracts, and web services. The contractor shall:

1. Develop and maintain in-house, COTS-based and open source-based applications that facilitate the creation of spatial data products.
2. Develop and maintain Web Mapping Services (WMS) for the distribution of spatial data.
3. Develop and maintain Web Feature Services (WFS) for the distribution of spatial data.
4. Develop and maintain Vector Tiles Services for the distribution of spatial data.
5. Design, develop, implement, and maintain software to create geographic reference and geographic equivalency files.
6. Provide design, development, implementation, maintenance, and enhancement support for both new and existing software and systems used to create and maintain the geographic codes utilized in the MAF/TIGER System and their relationships.
7. Modernize and enhance the GCAT System.
8. Evaluate technology usage and architecture of existing systems like GCAT and GPP and upgrade such systems to take advantage of newer technologies if needed.
9. Support provision of spatial and geographic products to customers in the USCB's Enterprise Data Lake (EDL), using tools and applications native to the EDL such as Postgres, Amazon Redshift, or Apache Hive.

10. Plan, develop, and provide an online service enabling public data users to bundle TIGER/Line GeoPackage Files into custom products.
11. Provide research and design support, agile and sprint support, develop and maintain software for all the regularly produced spatial and geographic products. A representative list as of 2025 is provided in the Current Environment (Attachment J-XX).

3.3.8.4 Geographic Areas and Partnership Operations

Geographic Areas and Partnership Operations support the numerous partnership operations that the USCB engages in to solicit and receive data from local partners, such as states, counties, incorporated places, minor civil divisions, and American Indian/Alaska Native areas. The data transmitted includes geographic boundaries, geographic attribute information, and address updates.

3.3.8.4.1 Ongoing Partnership Programs

The contractor shall:

1. Provide design, development, and ongoing operational assistance for production control systems, geographic data, and tools used by geographic partnership programs.
2. Aid with the identification and documentation of technical requirements for tracking, processing, and reporting systems used to process partner-supplied address lists and geospatial files.
3. Provide documentation of the above activities.

3.3.8.4.2 Geographic Catalog (GCAT) System

GEO requires the GCAT System to continue to add, edit, and display geographic data. The GCAT Graphical User Interface (GUI) is a web-based application providing the ability to add, edit, and remove geographic data of the United States and Territories. The GCAT System aids GEO in providing an independent repository of geographic entities, and their relationships, used to validate changes made in the MAF/TIGER System.

The contractor shall:

1. Maintain and enhance the web-based system that allows the USCB to add, edit, and/or remove geographic entity data within the United States and Territories.
2. Ensure that updates made through the system meet established business rules, criteria, and/or validation checks.
3. Ensure that the system has referential links to one or more of the USCB's geographic programs (e.g., Boundary and Annexation Survey).
4. Ensure that the system records transactions in an accessible repository.
5. Ensure that the system can display and edit geographic entity data in a web-based interface on census.gov.
6. Ensure the system supports multi-record updates for the Boundary and Annexation Survey operation.

7. Ensure the system has an interface for displaying and modifying legal values.
8. Ensure the system has interfaces for requesting and receiving help.
9. Ensure that the system links to Geographic Identification Coding Scheme (GICS) reports.
10. Ensure that the system integrates with the GPP System.
11. Ensure that all system data are stored in a relational database accessible to USCB staff.
12. Ensure that the system contains search functionality enabling USCB staff to search for geographic entity data as per requirements.
13. Ensure that the system complies with Section 508 Usability Standards as necessary.
14. Ensure that the system complies with all necessary security components and adhere to federal security standards, as well as USCB IT security requirements.
15. Ensure that the system complies with USCB IT Enterprise Architecture standards.
16. Ensure that the system meets the requirements for the risk profile and assessment under the USCB's Risk Management Framework Program.
17. Provide documentation of the above activities.
18. Support the modernization of the application and migrate it to the cloud.
19. Resolve security vulnerabilities that are discovered in GCAT.
20. Store GCAT source code and configuration files in the Census-approved Source Code Management (SCM) tool chosen by the USCB.
21. Collaborate with GCAT Product Owners to ensure that GCAT work is properly prioritized and delivered to the end user in the Production environment.
22. Maintain and provide support for GCAT in multiple environments, which includes Development, Test and Production. The support includes performance tuning, deployments, maintenance, and software releases.
23. Use a Census-approved Project Management tool to communicate the status of various work items that have been assigned for implementation.
24. Ensure that any new or updated GCAT code is properly tested and performant.
25. Provide support for GCAT Continuous Integration (CI) processes and Continuous Delivery (CD) processes.

3.3.8.4.3 Geographic Program Participant (GPP) System

GEO requires the GPP to continue to add, edit, and display program participant information, as well as generate reports and mailing extracts that GEO uses to mail geographic program materials. The GPP is a web-based application that provides the ability to add, edit, and remove program participant information of government contacts throughout the United States and its territories.

The contractor shall:

1. Maintain and enhance the web-based system that allows the USCB to add, edit, and remove contact information for governments within the United States and Territories.
2. Ensure that contact updates made through the system meet established business rules and/or validation checks.
3. Ensure that the system allows contacts to link to one or more of the USCB's geographic programs (e.g., Boundary and Annexation Survey).
4. Ensure that the system can link to PCSs to display and/or edit program participant information.
5. Ensure that the system can capture program participant information provided by participating governments from a web-based interface on www.census.gov.
6. Ensure that the system can produce canned reports as per the requirements provided by program areas.
7. Ensure that the system can produce custom-built reports as per the requirements provided by program areas.
8. Ensure the system enables an export functionality for all report types as per the requirements provided by program areas.
9. Ensure that all system data are stored in a relational database accessible to USCB staff.
10. Ensure that the system contains search functionality enabling USCB staff to search for contacts as per requirements.
11. Ensure that the system complies with Section 508 Usability Standards as necessary.
12. Ensure that the system complies with all necessary security components and adhere to federal security standards, as well as USCB IT security requirements.
13. Ensure that the system complies with USCB IT Enterprise Architecture standards.
14. Ensure that the system meets the requirements for the risk profile and assessment under the USCB's Risk Management Framework Program.
15. Ensure that the system integrates with the Geography Division Partner Portal (GDPP).
16. Provide documentation of the above activities.
17. Support the modernization of the application and migrate it to the cloud.
18. Resolve security vulnerabilities that are discovered in GPP.
19. Store GPP source code and configuration files in the Census-approved Source Code Management (SCM) tool chosen by the USCB.
20. Collaborate with GPP Product Owners to ensure that GCAT work is properly prioritized and delivered to the end user in the Production environment.
21. Maintain and provide support for GPP in multiple environments, which includes Development, Test and Production. The support includes performance tuning, deployments, maintenance, and software releases.

22. Use a Census-approved Project Management tool to communicate the status of various work items that have been assigned for implementation.
23. Ensure that any new or updated GPP code is properly tested and performant.
24. Provide support for GPP Continuous Integration (CI) processes and Continuous Delivery (CD) processes.

3.3.8.4.4 Geography Division Partner Portal (GDPP)

The GDPP is intended to provide external partners at tribal, state, and local governments secure access to entity-specific partnership program information, including geospatial data, historical program participation information provided by production control systems, access to applications such as the Secure Web Incoming Module (SWIM) (or SWIM replacement) and the Geographic Update Partnership Software (GUPS), contact data stored in the Geographic Program Participant (GPP) System, and a means to communicate directly with the USCB.

The contractor shall:

1. Maintain the web based GDPP system and recommend best optimization / migration paths or build solution paths and will receive SWG approval before implementing any new COTS into the GDPP implementation.
2. Maintain the retrieval of real time views of current program participation statuses, historical statuses of different geographic programs for an entity and display them in an easy-to-understand geographic display and tabular report by governmental level.
3. Establish and maintain interfaces, to facilitate exchange of data, between GDPP and the following systems and geographic programs:
    a) Geographic Program Participant (GPP) System
    b) Secure Web Incoming Module (SWIM) (or other Ingest Service)
    c) Secure Web Exchange Control System (SWECS)
    d) Geographic Update Partnership Software (GUPS)
    e) Production Control Systems (PCS's) of various geographic programs
    f) Boundary and Annexation Survey (BAS)
    g) School District Review Program (SDRP)
    h) Local Update of Census Addresses (LUCA)
    i) Redistricting Data Program (RDP)
    j) Spatial, Address, and Imagery Data Program (SAID)
    k) Federal Outreach Programs
4. Maintain a real time view of relevant content such as program participation statuses, historical data, contact information, etc.
5. Provide a link for users to access Title 13 data from other systems.
6. Allow download of forms, user manuals and other non-title 13 material.
7. Allow participants to securely and electronically sign any forms that are required (for example, title 13 confidentiality, product destruction, etc.).

8. Ensure GDPP communication platform maintains connection to the existing partnership program Shared Mailboxes (Mail-In Database) such as the GEO.GEOGRAPHY and GEO.BAS and other areas identified by the Technical Point of Contact (TPOC).
9. Ensure the GDPP maintains links to all the public and program-specific geographic materials/products for every geographic partnership program like maps, participant's guides, TIGER/Line and Partnership Shapefiles etc. and software/tools like the GUPS and the BAS Partnership tools that participants can use and download.
10. Ensure the portal maintains user-friendly and accessible interfaces so users can easily navigate among the different sections of the various geographic programs within the portal to find what they need.
11. Maintain the seamless user experience with simple, clean, and well-organized layouts.
12. Maintain compatible on multiple devices like computers (Window, Mac, etc.), laptop, tablets, and smartphones etc. and support different browsers like Microsoft Edge, IE, Mozilla Firefox, and Google Chrome etc.
13. Identify and develop communication methods to be sent and received by the staff of the individual geographic programs like BAS and SDRP to start in the Partner Portal. The communications should interface with any additional geographic programs identified. The communications shall include the incorporation of geographic program mail-in databases.
14. Ensure the GDPP maintains the communication platform for the participants and the USCB to send/receive any correspondence and to post any program notices/announcements.
15. Ensure and integrate access to the Web Geographic Update Partnership Software (GUPS) from the GEO Partner Portal (GPP) without (re)entering the username and password. Using the existing C-PASS credentials or similar solution on the Census Framework.
16. Maintain the secure login and authentication for users to create accounts and login to the Partner Portal. It should be a single sign on login that will allow users to navigate between modules without having to re-enter their login information Using the existing C-PASS credentials or similar solution on the Census Framework.
17. Maintain the GDPP build and deployment to be stored and operated in the Census cloud environment in a Cloud Managed Platform.
18. Maintain the documentation of the full implementation requirements for all environments (Dev, Test, Stage and Production).
19. Comply with all enterprise and bureau standards while implementing any solution.
20. Troubleshoot application deployments, recreate customer issues, and build proof of concept applications.
21. Write and interpret configuration scripts for customer environments to install programs, configure logging, and modify configuration files.
22. Monitor performance, availability, troubleshoot and addresses issues for applications and systems deployed in cloud environment.
23. Document system specifications, processes, and workflows.

24. Be responsible for hardware and software-related calls for all products that are sold or supported under the Order.
25. Give access to the public address range geocoder and other non-title 13 data.
26. Dashboard to display operation specific metrics.
27. Keep track of certain metrics (for example, when did they download the data, when did they try to do matching, how many times did they use the tools/ download the data etc.) as requested by program area.

3.3.8.4.5 JAGUAR System

The Joint Application Geographic Updater and Reviewer (JAGUAR) is a customized commercial-off-the-shelf (COTS) software that allows users the ability to view and edit spatial data in the MAF/TIGER System. While JAGUAR is a COTS product, Census is making enhancements to JAGUAR to make viewing and updating MAF/TIGER data more efficient, and correcting bugs that exist in the application. There are a limited number of new features being added, and efforts to improve the user experience and make JAGUAR more performant.

The Contractor shall:

1. Provide support to any identified effort required to maintain and update JAGUAR functionality as required for the Geographic programs.
2. Maintain the servers that are dedicated to JAGUAR processing, including: the JAGUAR User Interface servers, the JAGUAR Tile Servers, the JAGUAR Event Processor Servers.
3. Ensure that third party software applications required by JAGUAR, such as Apache Tomcat, and Apache ActiveMQ, integrate and synchronize well with JAGUAR.
4. Ensure that JAGUAR has sufficient Database Administrator support for the JAGUAR-specific data stored in an Oracle database.
5. Resolve security vulnerabilities that are discovered in JAGUAR.
6. Store JAGUAR-related source code and configuration files in the Census-approved Source Code Management (SCM) tool chosen by the USCB.
7. Collaborate with JAGUAR Product Owners to ensure that JAGUAR work is properly prioritized and delivered to the end user in the Production environment.
8. Maintain and provide support for JAGUAR in multiple environments, which includes Development JAGUAR, Test JAGUAR, Production JAGUAR, and Training JAGUAR. The support includes performance tuning, deployments, maintenance, and software releases.
9. Use a Census-approved Project Management tool to communicate the status of various work items that have been assigned for implementation.
10. Ensure that any new or updated JAGUAR code is properly tested and performant.
11. Provide support for JAGUAR Continuous Integration (CI) processes and Continuous Delivery (CD) processes.
12. Help with the upgrade of the Vaadin software component in JAGUAR from Version 8 to Version 24 or higher.

13. Update and test JAGUAR to ensure its compatibility with any AcquisConnect upgrades.

3.3.8.5 Map Production Software and Systems

The functional area of Map Production Software and Systems defines support that is provided to the GEO for the production of cartographic map products and services that are used in or result from other USCB operations. Regularly created cartographic applications and services are provided in the Current Environment (Attachment J-XX).

3.3.8.5.1 Cartographic Map Products

The Contractor shall:

1. Provide design, development, implementation, maintenance, and enhancement support for cartographic mapping software, databases, and systems.
2. Provide operational support for geospatial data processing in support of cartographic mapping: the automated generalized cartographic boundary file system, the creation of map files in graphics formats such as Adobe's Portable Document Format, the processing of Oracle spatial topology to calculate feature relationships, the generation and rendering of complex symbolization, and the calculation and transformation of map projections.
3. Provide documentation of the above activities.

3.3.8.5.2 Cartographic Services

The functional area of Geospatial and related COTS Software and System Management defines support that is provided to GEO for the use, implementation, administration, development, and management of Commercial Off the Shelf (COTS) software products and supporting system design. COTS Software and IT Systems Support provides GEO with support for:

- Managing COTS software deployed on servers in the USCB data center and in the cloud (AWS, etc.) and desktop computers to include VDI and thick client deployments.
- Assisting Government personnel with the monitoring and implementation of USCB IT Security Program Policies.
- Coordinating and managing server (hardware and software) resources with the USCB's IT Directorate.

This cumulative support applies to server (Windows and RHEL) and desktop software from the following COTS vendors: Esri, Safe Software (FME), Micro Focus, and Redwood Software.

The Contractor shall:

1. Provide support for COTS software applications installed within the GEO’s desktop and server environments (both on-prem and in the public cloud environments. This support includes:
    a) Performing software installations.

b) Interacting with COTS software vendors to obtain information related to new versions, and updates.

c) Troubleshooting software installation and performance problems, serving as a liaison between the IT Directorate and GEO subject matter experts and managers for matters related to COTS software.

d) Ensuring that COTS applications are installed, configured and utilized consistently with internal USCB and Department of Commerce IT security policies.

e) Manage user accounts for all GIS enterprise applications. Ensures membership roles are granted appropriately as established OIS policies dictate.

f) Maintain a secure, reliable infrastructure to deliver GIS products and services to both the Census user community and the public.

2. Provide Census enterprise-wide support for geospatial systems/software. This would include system architectural design, geodatabase management, integration of software into Census VDI and on Windows thick clients, and the installation, configuration, performance tuning, troubleshooting/problem resolution, and administration of COTS software.

3. Provide Census enterprise-wide COTS software support for the generation of spatial data, content, imagery, map and feature services, cache and vector tile, service publishing, web applications, story maps, web maps, and dashboards.

4. Provide after action reports (AAR) to document the root cause of system failures and actions taken to remediate and mitigate the risk of a recurrence.

5. Perform evaluation of new and emerging technologies to include support and/or participating in Analysis of Alternatives (AoA).

6. Transition systems as appropriate from the Census data center to the public cloud (e.g., AWS). Includes identifying candidate systems, performing system design, implementation and systems administration, and integration with DevSecOps pipeline for deployment of services, data, software updates, etc.

7. Provide support to manage the GEO's server, storage, and network resources in collaboration with the USCB's IT Directorate. This includes assisting with and performing software installations, assisting with troubleshooting and performance tuning, applying and/or monitoring the installation of software patches, implementing IT security standards and protocols, managing system permissions and access privileges (including SAMBA and other network shared storage), and assisting GEO with the planning for future server requirements.

8. Provide support to assist the GEO with responses to and actions resulting from the USCB's Office of Information Security (OIS) risk profiling, assessment, accreditation, and continuous monitoring processes. This includes the review of IT security scan results, remediation of identified vulnerabilities, documentation of false positive findings and risk acceptance, and ensuring the timely closure of associated Plan of Actions and Milestones (POA&M).

9. Support for Vector Tiles and Feature Framework Update System (FFUS) systems. Vector Tiles support the mapping components of EDDE. Vector tiles are packages of geographic data that contain information to draw geometries in a web map in an efficient

manner. A vector tile package is created for each summary level needed for a particular data release.

10. Provide support for Azure Monitor agent optimization and custom report and dashboard generation. Migrate Splunk dashboards to Azure reports.
11. Provide support for script development and process automation activities for the external management of COTS applications, application monitoring, and data deployments.

3.3.8.6 Geospatial Metadata Research and Support

Metadata is information about data. Like a library catalog record, metadata records document the who, what, when, where, how, and why of a data resource. Geospatial metadata describes maps, Geographic Information Systems (GIS) files, imagery, and other location-based data resources. The Federal Geographic Data Committee (FGDC) is tasked by Executive Order 12906 to enable access to National Spatial Data Infrastructure (NSDI) resources and by OMB Circular A-16 and the A-16 Supplemental Guidance to support the creation, management, and maintenance of the metadata required to fuel data discovery and access.

The contractor shall:

1. Support research to define, develop, and document current metadata creation processes for Geospatial products with specific skill requirements and tasks/potential deliverables detailed below.
2. Identify potential improvements and provide assessments and recommendations for current metadata processes, including automation of required metadata deliverables, quality control/quality assessments, and conformance, with particular emphasis on geospatial metadata.
3. Provide assessments and recommendations, as well as documentation of current internal USCB metadata production processes.
4. Provide assessment and recommendation of moving to alternate metadata standards.
5. Provide assessment and recommendation on the utilization of data creation conformance and QC tools, for example United States Geological Survey's (USGS) Metadata Editor suite (MDEditor) and National Oceanographic and Atmospheric Administration's (NOAA) Collection Metadata Editing Tool (CoMET).
6. Develop complete formal documentation for these assessments and recommendations.
7. Assist with implementation and programming of these assessments and recommendations.
8. Facilitate and participate in regular meetings and provide formalized meeting notes.
9. Provide formalized regular progress reports and task assessments.
10. Develop metadata subject matter expertise and provide regular opportunities for knowledge transfer to GEO/DITD staff and other contractors.
11. Provide guidance and recommendation for the creation of control system for the status and tracking of metadata.

3.3.8.7 Public Cloud Support (e.g. AWS, Azure)

The contractor shall:

1. Perform an assessment to determine what steps and software changes will be required to install and maintain MAF/TIGER applications in the cloud environment.
2. Develop documentation, such as a migration plan and guide, Installation guides/Release notes
3. Design, develop, and test identified software changes
4. Update the Existing MAF/TIGER application software design document for any applications migrated to the cloud will need to be updated.
5. Support and implement automation for installing and patching COTS and open-source software to the Census Gold AMI images. This will include Tomcat, Esri ArcGIS for Server, Esri Portal for ArcGIS, Esri Data Store, and geodatabases (to include file and enterprise DB geodatabase solutions)., etc.
6. Support the development of scripts and integration with CI/CD pipeline for supporting orchestration for the deployment of, and update to, software, services, and data.
7. Perform an analysis of alternatives for cloud-based database solutions. They shall further provide expert support in the development, implementation and management of applications using cloud-based database solutions.
8. Provide training to Government and contractor staff on the administration and management of adopted cloud-based solutions.
9. Build automated cloud testing for: scalability, performance, and security.
10. Analyze efficient ways and suggest best practices for implementing, monitoring, and deploying microservices in EKS /Serverless.
11. Establish a schedule and support the migration of MAF/TIGER applications to the cloud based agreed upon priorities for first migrating applications that don’t directly depend on the MAF/TIGER transaction database and then migrating MAF/TIGER transaction database-centric applications later.

3.3.8.7.1 Container Solution Adaptation and Implementation

USCB's modernization efforts involve adopting a cloud first strategy, leveraging open-source tools, building, and implementing secure solutions using latest development/trends in IT industry. Containers offer a unique opportunity to modernize our current legacy applications and develop new applications to take advantage of cloud services. The effort should result in an immutable infrastructure, ability to develop applications quickly, ability to quickly deploy, standardization, Agility to Scale to Demand, optimized compute resources, and improved security (transparency, security isolation, reduced attack surface, updates, consistent environment).

The contractor shall:

1. Develop, implement, and deploy a cloud container environment. This includes all development, all required coding and infrastructure support required to implement a full cloud container to support GSP products.

2. Maintain and update all existing programming already made to the CA like the GitLab Server and its communication pipeline to GitLab Runner.
3. Develop a Continuous Integration (CI) workflow or pipeline to create images and integrate with the Amazon Relational Database Service (RDS).
4. Develop an Elastic Container Registry (ECR) and Elastic Kubernetes Service (EKS) environment.
5. Develop Skopeo for the transfer of images between ECR and EKS.
6. Develop and ensure proper configuration of GitLab CI pipelines between Docker servers and the Amazon Web Services (AWS) Kubernetes Clusters.
7. Ensure this Cloud system connects and interfaces with all Geography systems and other identified Census systems.
8. Ensure proper configuration of the Nexus Repository for the upload and download of images using GitLab Runner.
9. Ensure proper communication and configuration of the Amazon Public Container Registry with the Nexus Repository.
10. Perform load tests of all systems and interfaces within the CA system.
11. Ensure the correct setup and configuration of all CA and Geography MAF/TIGER systems and interfaces are operable within it.
12. Ensure that any new software, development, deployments do not lose functionality with current Geography systems.
13. Research and implement solutions for automated testing and integration with a GitLab CI/CD pipeline for Red Hat Enterprise Linux (RHEL) processing (batch) and web-based applications.
14. Write scripts (yaml, Bash, PowerShell, Python, Ansible, etc.) for CI/CD pipeline in GitLab to incorporate testing scripts and report testing results and to validate deployments to database and middle tier environments as per requirements.
15. Perform the integration of MAF/TIGER applications with the CI/CD pipeline in GitLab to include writing yaml scripts and providing automation for software deployments to test, stage, and production environments.
16. Develop and implement Kubernetes/EKS, ECS, Docker and containerized solution architectures with advanced engineering concepts around security, segmentation, vulnerability and image scanning, and integration.
17. Automate deployment of containers to a Kubernetes based environment like AWS-EKS using GitLab CI/CD, Terraform, and Helm.
18. Create applications using Docker containers and Kubernetes, perform feasibility study, and port MAF/TIGER applications to containers.
19. Prepare secure baselines and configure EKS, EFS, ELB, ECR, and other AWS cloud tools.
20. Assure container runtime security and Cloud-Native Application Protection (CNAP)
21. Assure container rehydration monitoring and vulnerability management.

22. Coordinate with application teams to drive security tooling into the pipelines.
23. Be responsible for Packaging and distribution of secure builds for virtual machines and base-containers.
24. Build comprehensive security controls to enforce bureau policies.
25. Troubleshoot application deployments, recreate customer issues, and build proof of concept applications.
26. Write and interpret configuration scripts for customer environments to install programs, configure logging, and modify configuration files.
27. Monitor performance, availability, troubleshoot and addresses issues for applications and systems deployed in cloud environment.
28. Document system specifications, processes, and workflows.
29. Comply with all enterprise and bureau standards while implementing any solution.
30. Comply with USCB standards in tools/software usage.
31. Document the full implementation requirements for all environments (Dev, Test, Stage and Production).

3.3.8.8 Support for GitLab DevSecOps, CI/CD Integration, Automated Software Deployments, and Automated Testing

The contractor shall:

1. Adapt automated security vulnerability and configuration scanning for open-source components and commercial packages.
2. Focus on identifying and remediating known security vulnerabilities
3. Implement Strong Version Control on All Code and Components
4. Integrate Security Testing within the CI/CD Pipeline adapting Static application security testing (SAST) and dynamic application security testing (DAST), secret scanning, dependency scanning, container scanning, IAC scanning, API security, and fuzz testing.
5. Integrate testing as part of CI/CD pipeline.
6. Train Government and contractor developers on secure coding
7. Adopt an immutable infrastructure.
8. Research and implement open source and COTS software solutions for automated testing and integrate with a GitLab CI/CD pipeline for Red Hat Enterprise Linux (RHEL) processing (batch) and web-based applications.
9. Write custom scripts (yaml, shell, etc.) for CI/CD pipeline in GitLab to incorporate testing scripts and report testing results and to validate database and middle tier per requirements.
10. Perform the integration of MAF/TIGER applications with the CI/CD pipeline in GitLab to include writing yaml scripts and providing automation for software deployments to test, stage, and production environments.

11. Research and implement open source and COTS solutions assuring alignment with industry best practices for programming and IT security compliance.
12. Provide training to Government and contactor staff on performing on all DevSecOps related automation activities.
13. Participate as an SME/Developer in DevSecOps Implementation project.
14. Research, implement, administer, and manage Open Source DevSecOps tools in plan, build, test, deploy, monitor, and operate stages of the pipeline.
15. Write scripts (yaml, Bash, PowerShell, Python, Ansible, etc.) for CI/CD pipeline in GitLab to incorporate testing scripts and report testing results and to validate deployments to database and middle tier environments as per requirements.
16. Perform the integration of MAF/TIGER applications with the CI/CD pipeline in GitLab to include writing yaml scripts and providing automation for software deployments to test, stage, and production environments.
17. For all new software releases using the DevSecOps pipeline, documentation is due at the time of software release.

3.3.9 2030 Census Geographic Program Support

The scope of this functional area is to provide the Geographic Support Program support for 2030 Census related activities. These geographic activities, while similar to other activities outlined above, are required to be identified and tracked separately as they are identified throughout the lifecycle of the 2030 Census and include the following.

1. **Geographic Delineation:** The contractor shall support geographic delineation and updates for the collection geographic area boundaries required for the 2030 Census, including all 2030 Census testing activities.
2. **Address Products and Services:** The contractor shall support the creation of address products and services, the processing of address submissions from internal and external operations, and the update of the MAF with the resulting actions from all 2030 Census operations.
3. **Spatial Products and Services:** The contractor shall support the Decennial map production and plotting, decennial spatial data deliveries, collection geographic reference files, and any other spatial products and services, including ones in support of Geographic Area Program. This includes supporting the implementation of mapping services in Enterprise Data Lake (EDL).
4. **Census Geocoder**: The contractor shall support the 2030 design, development, testing, and implement enhancements to the geocoding services within the EDL.
5. **2030 Applications Support**

3.3.9.1 2030 Applications

3.3.9.1.1 2030 Production Control System's (PCS)

The contractor shall:

1. Support the development of PCSs for 2030 Census programs and operations, such as the Local Update of Census Addresses, New Construction, Participant Statistical Areas Program, Redistricting Data Program, and others require a PCS to assist with storing, tracking, and reporting on day-to-day updates. The PCS is integral in keeping staff aware of daily information regarding partnership participation, submissions received, communications, Feedback, and Appeals.
2. Leverage existing code from other PCS such as BAS or SDRP as much as possible to meet expected production dates, minimize future maintenance, and ensure all systems are in sync. Please note the PCS will need to communicate with the other applications, such as the Partner Portal, SWIM (or new ingest service), GUPS Web, Intelligence Database, Matching, Validation, and any other future systems the GEO deems necessary to fulfill the scope of the 2030 Census.

3.3.9.1.2 2030 WEB Enabled Batch Control Systems (WEBCS)

The contractor shall:

1. Support Workflow Control and Management for managing, controlling, and reporting for batch processing operations that interact with the MAF/TIGER System and for large GEO operations that have multiple project phases. This is accomplished using both COTS software as well as custom-developed applications.
2. Design, develop, implement, unit test, support, maintain and enhance automated production workflow control and reporting systems. This includes file management systems, WEB enabled Batch Control systems (WEBCS), running applications for batch creation of products such as address, spatial and Local Update of Census Addresses (LUCA) and Redistricting Data products.

3.3.9.1.3 GDPP for 2030 Census

The GDPP is intended to provide external partners at tribal, state, and local governments secure access to entity-specific partnership program information, including geospatial data, historical program participation information provided by production control systems, access to applications such as the Secure Web Incoming Module (SWIM) (or Data Ingest and Collection for the Enterprise [DICE] replacement) and the Geographic Update Partnership Software (GUPS), contact data stored in the Geographic Program Participant (GPP) System, and a means to communicate directly with the USCB. The GDPP, along with the section above for the Geographic Support Program, will be utilized for 2030 Census Geographic Operations such as LUCA.

The contractor shall:

1. Support the continued development of the GEO Partner Portal (GDPP) for 2030 Census programs and operations

3.3.9.1.4 LUCA Module and Interface for GDPP

The LUCA operation requires a module in the GDPP to interact with government participants in the operation. The following are the minimum requirements:

1. Allow participants to securely and electronically sign any forms that are required (for example, title 13 confidentiality, product destruction, etc.).

2. Provide updates from PCS to the partner portal.
3. Allow communications between census staff and participants.
4. Allow download of forms, user manuals and other non-title 13 material.
5. After policy and security approval, allow participants to download title 13 material if the user is certified.
6. After policy and security approval, give access to the address matching service (title 13), allow them to download the outputs, and allow them to upload submissions.
7. Give access to the public address range geocoder and other non-title 13 data.
8. Dashboard to display operation specific metrics (for example, how many MAF units in an entity).
9. Keep track of certain metrics (for example, when did they download the data, when did they try to do matching, how many times did they use the tools/ download the data etc.).
10. Dashboard to display operation specific metrics.

3.3.9.1.5 Matching and Coding System (MaCS) for the 2030 Census

MaCS is an interactive web interface that encompasses address modules that are built for the needs of specific operations for the 2030 Census as well as activities including Census Tests. All modules have one or more of these functionalities: user account management, workload management, clerical data entry, text search of MAF addresses, address ranges, and other references, map interface, reports, and quality control. There are also training datasets that are typically needed.

The contractor shall:

1. Design, develop, test and implement future enhancements to MaCS that could include:
   a) MaCS In-Office Address Canvassing Group Quarters and Transitory Locations (IOAC GQ/TL) - The MaCS IOAC-GQ/TL module enables staff to research, call, verify and validate GQ/TL addresses and data from the MAF to determine whether the address is correct or requires modifications. In addition, it enables staff to add newly identified GQs when finding information from research or phone interview.
   b) MaCS Virtual Canvassing - The MaCS Virtual Canvassing module enables staff to review the identified areas using various reference sources and update the addresses and assign an action code to each address to denote a change or lack of change to a block count unit (BCU).
   c) MaCS Count Question Resolution (CQR) - The MaCS CQR application enables CQR staff to perform address matching on CQR cases submitted by governmental entities. Addresses that do not match are then researched by CQR researchers where they attempt to find MAF units that represent the addresses.

3.3.9.1.6 MaCS Address Geocode Enhancement (MAGE)

MAGE is an environment within EDL supporting clerical Non-ID and LUCA Address Validation (LAV) operations. For example, all non-ID records that do not get auto-matched and resolved during Central Decennial Address Control (CDAC) operations feed into MAGE for additional address range geocoding, header coding, and address clerical research operations within the MaCS-based Decennial Address Coding System (DACS). In addition, DACS serves as the tracking and control system of all transactions within MAGE and its incoming and outgoing data transactions with CDAC. MAGE will require contractor support such as updates and maintenance.

The contractor shall:

1. Support the development of MAGE

3.3.9.1.7 Decennial Address Coding System (DACS) for the 2030 Census

DACS is a MaCS module being designed and developed as a cloud-based application to support decennial Non-ID and LUCA Address Validation (LAV) address matching operations. For non-ID, it will accept respondent records that do not have a pre-assigned MAFID. Respondent records with no pre-assigned MAFID come from the following sources:

- Internet Self-Response (ISR)
- Census Questionnaire Assistance (CQA)
- Mobile Questionnaire Assistance (MQA)
- In-Field Enumeration (IFE)
- Usual Home Elsewhere (UHE)

The goal of Non-ID in DACS is to assign a MAFID to each of the respondent records via Office Based Address Verification and/or DACS Clerical Coding. For the decennial activities, DACS will interface with the Centralized Decennial Address Control (CDAC) in EDL. In addition, DACS will incorporate LAV operations into its system due to its operational similarities with Non-ID (LUCA is not part of the 2026 Census Test).

The contractor shall:

1. Support the development, updates, and maintenance for DACS

3.3.9.1.8 Support for DevSecOps CI/CD Integration, Automated Software Deployments, and Automated Testing.

The contractor shall:

1. Evaluate software requirements to develop/update test cases and other test reports for the myriad of 2030 Census Operations (as outlined in Section 6.3) interactive and batch software that comprise the MAF/TIGER System according to established standards and guidelines and industry best practices for use in independent testing and validation efforts.
2. Create test cases, develop automated test scripts (currently using GitLab and

Katalon), and conduct independent testing according to established guidelines and industry best practices for each 2030 Census Geographic Program Support (as outlined in Section 6.3) MAF/TIGER System application (Red Hat Enterprise Linux (RHEL) processing (batch) and web-based applications) that will undergo development (new application, redesign, or Cloud transformation) within the established and documented corresponding software application project schedule as agreed upon by the U.S. USCB.

3. Write custom scripts (yaml, shell, etc.) according to industry best practices for CI/CD pipeline in GitLab to incorporate independent software validation automated testing scripts and report software validation testing results for each 2030 Census Geographic Program Support (as outlined in Section 6.3) MAF/TIGER System application (Red Hat Enterprise Linux (RHEL) processing (batch) and web-based applications) that will undergo development (new application, redesign, or Cloud transformation) within the established and documented project schedule as agreed upon by the U.S. USCB.
4. Be a member of the DevSecOps team for which the test cases, automated test scripts, and testing are needed. This includes testing in development environment during Agile Sprints and in the Test Environment when software is released as agreed upon by the U.S. USCB.
5. Ensure any new DevSecOps application (new, redesign, or Cloud transition) shall have test cases and independent automated test scripts built and integrated into the CI/CD pipeline within the established and documented project schedule as agreed upon by the U.S. USCB.
6. Coordinate the integration of the automated test scripts for independent testing and validation into the CI/CD pipeline with all required teams and within the established and documented corresponding software application's schedule as agreed upon by the U.S. USCB.
7. Provide training to government and contactor staff on performing all related interactive and DevSecOps automated testing activities.
8. Research, implement, and manage open source and COTS solutions assuring alignment with industry best practices for automated interactive and batch DevSecOps independent testing.

3.3.9.1.9 2030 Address Management Tool (AMT)

The Address Management Tools consist of the following components:

- Quality Check Service – a software API that checks the quality of a participant's address list and reports the results
- Wizard – a UI and software API that controls processing flow
- Address Matching Service (AMS) - a software API that matches a participant's address list against the Census address list
- AMT storage for T13 data – a Cloud-based AWS data bucket that stores Title-13 data

The contractor shall:

1. Support the ongoing maintenance of the 2030 Census Address Management Tools, including the integration with the 2030 Census Internet Self Response (ISR).

3.3.9.2 Centralized Decennial Address Control (CDAC) Project Support

The scope of this functional area is to provide the MTDB System support for research improvements, development and implementation for Census tests and the 2030 Census for geographic program management and processing as well as ability to get support to help with documentation of this research. The research and innovation would include the following: storage and retrieval spatial data, processing of spatial and address data, and new tools and technologies.

Additional activities under this task area include the following:

- New production control systems
- Operation specific control systems
- Documentation and training
- GIS systems support
- Processing, workflow, and scheduling
- Reports
- Address Viewer
- System Integration with other 2030 Census Applications and databases.

3.3.9.2.1 Support Cloud Applications and Products

The contractor shall:

1. Support application development in the USCB's Enterprise Data Lake (EDL).
2. Utilize cloud and Big Data technologies to support setup and management of the Centralized Decennial Address Control database in the EDL.
3. Utilize, Amazon Redshift, Postgres, API Gateway, Lambda functions, Iceberg Tables, Event Bridge Notifications, or other EDL tools to extract, process, match, and analyze data, to include 2020 Census Collection Event data, 2020 Census data, other pertinent Administrative Record data, and information on 2030 Census Research Prototype systems.
4. Support provisioning of MAF/TIGER System products in the EDL to lessen product duplication and provide a central location for access of products by multiple customers.
5. Provide a written justification and rationale for new products or services needed if identified through contractor research.

3.3.9.2.2 Support Real-Time Matching

The contractor shall:

1. Design, develop, test application software to support near real-time address matching and processing capability for the decennial address universe. This real-time matching will endeavor to drastically reduce the turn-around time for matching to the address repository from months to minutes or seconds.
2. Provide design, development, migration and testing support to move the real-time matching capability to the EDL using PostgreSQL or equivalent.
3. Support the work to estimate peak workloads for address adds and updates during decennial operations
4. Support Collection of metrics on real-time matching performance
5. Support development of a final analysis and lessons learned regarding the feasibility of real-time matching, as well as documentation of issues that need to be considered in the CDAC design.

3.3.9.2.3 Support Development of CDAC

In support of the CDAC that is part of the 2030 Research Program including production for 2026 Census Test, 2028 Dress Rehearsal, and 2030 Census, the contractor shall:

1. Support development and maintenance of a concept of operations for the CDAC throughout the lifecycle Solution architecture for the CDAC.
2. Support creation and maintenance of data and process workflow documentation for CDAC throughout the lifecycle.
3. Support documentation of the physical database design of PostgreSQL CDAC database.
4. Support design, development, and implementation of scripts for extraction of address data from MTDB.
5. Support transformations of extracted address data for loading into PostgreSQL database and CDAC database structures.
6. Support design, development, and implementation of scripts for batch loading of address data into CDAC database.
7. Support generation of bulk address universe files from CDAC for transmission to MOJO through a survey input file (SIF).
8. Support design, development and implementation of address update software to update the CDAC based on address transactions from external operations.
9. Support design, development and implementation of Address Ingest services to accept a single or multi record streams.
10. Support design, development and implementation of business rules and quality control checks into Address update process.
11. Support design, development and implementation of Real Time Non-ID Processing (RTNP) web service used to support the Internet Self Response instrument, including standardization, header coding, matching and geocoding.
12. Support design, development and implementation of all decennial system interfacing per stakeholder requirements or change requests.

13. Support design, development and implementation of near real-time matching and standardization services to support non-ID addresses.
14. Support management of transmission of addresses to and from the operations that geocode and validate non-ID addresses.
15. Support ingesting and storing Collection Events or equivalent in CDAC database
16. Support generation of mail address lines for printing (Address Composition).
17. Support ingest of sampling variables and additional data items from DSSD.
18. Support Sampling calculations.
19. Support assignment of IDs to new addresses.
20. Support transmission of addresses to MAF/TIGER Systems for geocoding or geocode validation.
21. Support generation of individual transaction records (adds, updates, etc.) to send to MOJO through a SIF.
22. Support calculation of Final Collection Universe for 2030 Census and Census tests.
23. Support integration with DevOps pipeline for deployment of services, data, software updates, etc.
24. Support development of test decks for testing.
25. Support design, development and implementation of tallies and reports.
26. Support CDAC interfaces integration with In-office Enumeration, Near Real Time Processing (NRTP), Quality (QU) Assessment (A) in near real (N) time (TM) (QUANTM) and other Decennial Systems as needed.

3.3.9.2.4 Support for 2030 Census Design Selection Projects and Interfaces

The contractor shall:

1. Provide software design and development support for select 2030 Census Design Selection projects, specifically as related to interfaces with the CDAC, migration to EDL, and utilization of EDL tools and technology.
2. Support documentation of interfaces with other decennial systems
3. Design, develop, implement, and test software to interface with other decennial systems
4. Support utilization of EDL tools and cloud technology as needed to support decennial interfaces in EDL.

4.0 Staffing and Key Personnel

The Contractor's Key Personnel will be identified for this Order. The Contractor shall assign and identify the Key Personnel for the duration of the Order period of performance and provide resumes for these personnel. The Contractor shall not divert or otherwise replace any key personnel without the written consent of the CO. The government may modify the Order to add or delete key personnel at the request of the Contractor or government.

Substitution or replacement of the Key Personnel should be avoided to the extent practicable within the first ninety (90) days after Order award.

All requests for approval of changes hereunder must be in writing, via email, and provide a detailed explanation of circumstances necessitating the proposed change. Request for changes should be made whenever the need is identified. In addition to the resume, the request must also provide:

1. A comparison of skills and qualifications to those set forth in the accepted resume proposed for substitution.
2. A signed employee procurement integrity agreement.
3. Number of hours the Contractor shall provide at his/her own expense to train the proposed replacement.
4. Any other information requested by the Contracting Officer to reach a decision.

Any personnel the Contractor offers as a substitute shall possess experience and qualifications equal to or better than the personnel to be replaced. Requests to substitute personnel shall be approved by the CO and Task Lead. All requests for approval of substitutions in personnel shall be submitted in writing to the CO 14 calendar days prior to the requested change date for any change in Key Personnel.

Role/Title	Decennial BPA LCAT	Full Time Equivalents (FTEs)	Requirements (Qualifications)	Task Area(s)
Program Manager	Program Manager - SME	**1**	**Experience**: Provides technical / management leadership on major tasks or technology assignments. Establishes goals and plans that meet project objectives. Is experienced in the geospatial industry and possesses expert technical knowledge. Directs and controls activities for a client, having overall responsibility for financial management, methods, and staffing to ensure that technical requirements are met. Interactions involve client negotiations and interfacing with senior management. Decision-making and domain knowledge may have a critical impact on overall project implementation. May supervise others. **Years of Experience:** 15+ Years **Preferred Requirements:** Project Management Professional (PMP) Certification	Task Area 1 & 2

Role/Title	Decennial BPA LCAT	Full Time Equivalents (FTEs)	Requirements (Qualifications)	Task Area(s)
Systems Engineer	Systems Engineer – SME	1	**Experience:** Provides technical leadership for supporting the design, implementation, and maintenance of IT systems. Responsibilities include analyzing business requirements and translating into technical solutions that are aligned to the business / mission needs; employing appropriate architectural frameworks and principles; driving the implementation and utilization of scaled agile methodologies; developing, updating, and maintaining system design solutions and documentation; supporting quality assurance processes for adherence to standards and best practices. Is experienced in the geospatial industry **Years of Experience:** 15+ Years **Preferred Requirements:** Expert Systems Engineering Professional (ESEP)	Task Areas 2 & 3

5.0 Technical Instructions

Technical instructions to perform effort under the Order PWS may be given by means of TIs issued in numerical sequence. TIs may be provided to the Contracting Office with the initial Order Requirements Package, or with option exercise requests to describe the effort to be performed. A TI may be modified, cancelled, or superseded any time by issuance of a new TI. TIs may be issued during the course of Order performance to provide technical direction that may more closely reflect new information or changed priorities within the TO PWS.

Each TI will establish the effort to be expended for its performance and may include the number of manhours, travel, direct material, and/or other direct costs. The TI shall identify the PWS paragraph number(s) under which the TI effort is covered. The TI shall clearly be tied to the applicable TO SLIN (if applicable) under which the effort is to be performed. If the TI is used to identify the price of that individual effort, in no event shall the total number of labor hours and price issued under a series of TIs exceed those labor hours and price set forth in section B of the specific Order for a particular SLIN.

5.1 Process for Defining Annual Requirements, Work Plans, and Deliverables

Through this Order, GEO will require the contractor's assistance, skills, and expertise to accomplish the stated objectives in alignment with the Task Areas defined in the PWS. The Technical Instructions (TI) will detail work assignments, specify required work products, deliverables, and set schedule objectives.

The table below outlines the notional schedule for the award of TIs:

Day 1	GEO issues statement of work for new task
Day 21	GEOTAM Contractor provides solution for task (technical and cost)
Day 30	Negotiations complete
Day 40	Funding is provided to add a new CLIN w/Authorization to Proceed
Day 45	Performance begins

The Contractor shall:

1. Provide staff to support the business management of all TIs issued under this contract.
2. Draft technical solutions and cost estimates within schedule to meet the needs of the GEO.
3. Establish and document comprehensive activity schedules and milestones.
4. Establish reporting requirements and describe cost and schedule reporting mechanisms. These requirements include customized reporting on planned or historical costs and activities as needed for GEO to meet requests from the Department of Commerce (DOC), the Government Accountability Office (GAO), Congressional Oversight bodies, and other Government entities.
5. Maintain auditable records and contract files for all business transactions under this contract.
6. Respond to ad hoc requests for information from internal and external stakeholders.
7. Provide cost, schedule, and progress reporting for all work in progress.
8. Maintain accurate accounting and billing across all project codes, CLINs, and SLINs under this contract.

6.0 Period of Performance

The GEOTAM Order comprises a one-year base performance period with seven, one-year option periods.

Contract Year	Start Date (estimated)	End Date
Base Period	09/22/2026	09/21/2027
Option Year 1	09/22/2027	09/21/2028
Option Year 2	09/22/2028	09/21/2029
Option Year 3	09/22/2029	09/21/2030

Contract Year	Start Date (estimated)	End Date
Option Year 4	09/22/2030	09/21/2031
Option Year 5	09/22/2031	09/21/2032
Option Year 6	09/22/2032	09/21/2033
Option Year 7	09/22/2033	09/21/2034

7.0 Place of Performance

Services shall be performed at the USCB located at 4600 Silver Hill Road, Suitland, MD, 20746. Services may be performed at other USCB facilities and remote locations not directly managed by the USCB only with approval of the COR, and subject to any other approvals as required by USCB policy and described below.

The COR may authorize the performance of services under this BPA at facilities not managed by the USCB provided those services do not involve access to or use of: USCB data protected under Title 13 U.S.C.; Federal Tax Information protected under Title 26. U.S.C.; any system that stores or processes data protected under Title 13 or Title 26; or the USCB's corporate information technology (IT) network. Contractors are, at all times, prohibited from accessing any of these data or assets from non-authorized locations.

The COR may authorize performance of services under this BPA (including services that require electronic access to information protected by Title 13 U.S.C. and if authorized by the Internal Revenue Service information protected by Title 26 U.S.C.) at the contractor's individual place of residence. To facilitate performance of these services, access to USCB IT resources may, subject to Division or Office Chief approval, be granted via a Virtual Private Network (VPN) enabled USCB-issued Laptop or through the Virtual Desktop Infrastructure (VDI).

Except as described above, the performance of any services at locations not managed by the USCB that involve access to or use of: data protected under Title 13 U.S.C.; Federal Tax Information protected under Title 26 U.S.C.; any system that stores or processes data protected by Title 13 or Title 26; or the USCB's corporate IT network may be subject to the following additional approvals:

- Review and approval of the physical security controls at the remote location by the USCB's Office of Security. This review may include an on-site inspection.
- Review and approval of the IT security controls in place at the remote location by the USCB's Office of Information Security. This review may involve the authorization of any systems that will handle and process the USCB's data, or that will connect to the USCB's network.
- Review and approval by the Internal Revenue Service if the services to be performed involve access to or use of Federal Tax Information protected under Title 26 U.S.C.

- Review and approval by the USCB's Policy Coordination Office and/or Data Stewardship Executive Policy Committee.

Contractors are not authorized to work remotely with Federal Tax Information (FTI) protected by Title 26

U.S.C. unless executed by federal government contingency planning due to uncertain events and approval from the Internal Revenue Service (IRS). If allowed to work remotely with FTI, each vendor contract must contain language for safeguarding federal tax information as mentioned in IRS Publication 1075, Exhibit 7 (https://www.irs.gov/pub/irs-pdf/p1075.pdf).

Contractors may only perform services for the USCB from approved locations. If authorized to work remotely, contractors must still adhere to all applicable USCB Data Stewardship and IT Security Policies and complete all required training. The COR is responsible for ensuring the required training is completed for every contractor authorized to work remotely, and the contractor's access to USCB IT resources and data assets will be terminated for non-compliance.

8.0 Hours of Operation

Contractor staff shall provide support during the core hours from 6:00 am – 7:00 pm EST, Monday through Friday. Key Personnel shall be scheduled during hours 9:00 am – 5:30 pm EST, Monday through Friday unless prior authorization or notification of absence has been given. Support is deemed after-hours and/or on-call, before 6:00 am and after 7:00 pm, Monday through Friday. To support emergency operations, benchmark support, maintenance, and testing schedules, and to assist with GEO's effort of almost 100% operational uptime, the Contractor personnel shall be on-call before and after normal hours of operation, including weekends (Saturday and Sunday) and holidays. Regular maintenance such as hardware parts, software patches and update support will be planned and scheduled after normal hours of operation, including weekends (Saturday and Sunday) and holidays as requested by the COR. The contractor will provide after-hours support as needed.

The contractor is responsible for adhering to all established service levels provided in the Service Level Agreements (Attachment J-XX) and shall adhere to the timeframes for issue resolution during core hours from 6:00 am – 7:00 pm

9.0 Government Furnished Resources

The Contractor shall be responsible for safeguarding all government equipment, information and property provided for contractor use. The Contractor shall not use GFP/E/I for any purposes other than official government business as performed under this Order. At the close of each workday, government equipment and materials shall be secured.

Contractor personnel working on government sites will be provided with standard business equipment including all or some of the following: desk, chair, phone, computer, and access to office equipment such as printers, copiers, fax, etc. as appropriate.

10.0 Travel

When anticipated, GEO will identify travel requirements required to support Order objectives in addition to providing the government with a travel cost estimate at the beginning of each period of performance. Reimbursement of local commuting expenses is not allowed for travel inside the Washington, DC metropolitan area. Local travel is defined as within 50 miles.
The Contractor shall adhere by the following travel stipulations:

1. All travel shall be billed in accordance with the General Services Administration (GSA) Federal Travel Regulation.
2. Travel shall not be taken without prior, written approval of the COR and Government technical lead using the Travel Authorization form.
3. Verifying and validating all travel receipts and Government approvals are included with travel invoices and delivered within five (5) days from return of travel.

Local travel to meetings, conferences or other official business covered under this Order is not reimbursable as a direct charge. Local travel is defined as within 50 miles.
If travel is required, the Government requires the Contractor to obtain prior written approval, using the Travel Authorization Form (TAR) (Attachment J-XX), from the Government Team Lead and designated COR to attend meetings, conferences or conduct other official business covered under this TO. The Contractor shall submit requests for travel, using the Travel Authorization Form (TAR), to the Government Team Lead and COR at least 14 business days in advance of the travel. The Contractor shall submit a Trip Report according to the timing identified in the Deliverables Section. Authorized travel is reimbursable in accordance with the Order and in alignment with (IAW) the Federal Travel Regulation (FTR) and FAR Part 31.205-46 – Travel Costs. The Contractor shall ensure the amount is consistent with the not-to-exceed (NTE) amount specified for this contract line item (CLIN). Travel shall be in accordance with GSA approved rates – i.e. Per Diem, Lodging. A summary of Travel expenses which includes original or legible copies of receipts as prescribed in the FTR shall be submitted with the monthly invoice for the travel occurred.
The FTR may be located and downloaded from: [Federal travel regulation | GSA].

Any burden added to the travel cost will be allowed only as defined in the Contractor's standard accounting practice or disclosure agreement.

11.0 Security

11.1 Purpose

The U.S. USCB (USCB) must provide information security for the information systems that support the operations and assets of the agency, including those provided or managed by another agency, contractor, or other source. The Federal Information Security Modernization Act of 2014 (FISMA) describes Federal agency security responsibilities as including "information systems used or operated by an agency or by a contractor of an agency or other organization on behalf of an agency." This includes services which are either fully or partially provided; including other agency hosted, outsourced, and cloud computing solutions. FISMA has a somewhat broader applicability than prior security law and applies to both information and information systems used by the agency, contractors, and other organizations and sources. Agency information security programs apply to all organizations (sources) which possess or use Federal information – or which operate, use, or have access to Federal information systems (whether automated or manual) – on behalf of a federal agency, information systems used or operated by an agency or other organization on behalf of an agency.

This document applies to all Agency contracts in which USCB information, Title 13, Title 26 and or Controlled Unclassified Information (CUI) is stored, generated, transmitted or exchanged by an USCB contractor, subcontractor or third-party, or on behalf of any of these entities regardless of format. The regulations in this document also pertain to information residing on an USCB or a non-USCB system in Order for the contractor, subcontractor or third party to perform their contractual obligations to USCB, standing in lieu of USCB or acting on USCB's behalf.

The contractor shall leverage all applicable enterprise IT services available from the Office of the Chief Information Officer (OCIO) and the Office of the Chief Information Security Officer (OCISO). The goal of enterprise services is to maintain enterprise protection and visibility of all agency IT systems. The agency expects a lower cost of ownership, less complexity, lower level of maintenance effort and overall cost savings over the life of the contract. The services include:

- Enterprise audit log aggregation
- 24/7 Security Monitoring and Incident Response
- Enterprise Patch and Configuration Management
- Enterprise Vulnerability Scanning
- Annual and Ad Hoc Penetration Testing
- Annual and Ad Hoc Tabletop Exercises for Incident Response
- Enterprise Cybersecurity Continuous Monitoring

All requirements identified or referenced in this document shall be interpreted and implemented implicitly for all system components unless otherwise directed by the USCB.

All requirements identified or referenced in this document shall be implemented at the time of contract award unless otherwise directed by the government.

All requirements identified or referenced in this document shall apply to all contractors and subcontractors that will have access to CUI, collect or maintain CUI on behalf of the agency, operate federal information systems, including contractor information systems operated on behalf of the agency, to collect, process, store, or transmit CUI, Title 13 and Title 26 information.

All physical and/or logical access to the IT system or supporting facilities deemed necessary by the USCB shall be granted to the USCB or its designated representative(s) by the Contractor under Federal Acquisition Regulation (FAR) Part 52.246 (Contractor Inspection Requirements).

11.2 Policies and Regulations

Contractors entering into an agreement for services to the USCB or its Federal customers shall be contractually subject to all USCB and Federal IT Security standards, policies, and reporting requirements. The contractor shall meet and comply with all USCB IT Security Policies, USCB and NIST guidelines, other Government-wide laws and regulations for protection and security of Information Technology.

Contractors are required to comply with the USCB, FIPS, and NIST, Federal requirements outlined below (or successor documents):

- USCB Information Technology Security Program Policy ITSPP (current version).
- Federal Information Security Modernization Act of 2014, as amended.
- Internal Revenue Service Publication 1075
- Clinger-Cohen Act of 1996 also known as the “Information Technology Management Reform Act of 1996,” as amended.
- Privacy Act of 1974 (5 U.S.C. § 552a), as amended.
- Federal Information Technology Acquisition Reform Act (FITARA) of 2014, as amended.
- Chief Financial Officers Act of 1990 (Public Law 101–576), as amended.
- Cyber Supply Chain Management and Transparency Act of 2014.
- Trade Agreements Act (TAA) of 1979 as amended.
- Homeland Security Presidential Directive (HSPD-12), “Policy for a Common Identification Standard for Federal Employees and Contractors,” as amended.
- Office of Management and Budget (OMB) Circular A-130, “Management of Federal Information Resources” as amended.
- FIPS PUB 199, “Standards for Security Categorization of Federal Information and Information Systems,” as amended.
- FIPS PUB 200, “Minimum Security Requirements for Federal Information and Information Systems,” as amended.
- FIPS PUB 140-2, “Security Requirements for Cryptographic Modules,” as amended.
- NIST Special Publication 800-18, “Guide for Developing Security Plans for Federal Information Systems,” as amended.
- NIST Special Publication 800-30-1, “Guide for Conducting Risk Assessments,” as amended.
- NIST Special Publication 800-34-1, “Contingency Planning Guide for Information Technology Systems,” as amended.
- NIST Special Publication 800-37-2, “Guide for Applying the Risk Management Framework for Information Systems and Organizations: A System Life Cycle Approach for Security and Privacy,” as amended.
- NIST Special Publication 800-47, “Security Guide for Interconnecting Information Technology Systems,” as amended.
- NIST Special Publication 800-53-4, “Security and Privacy Controls for Federal Information Systems and Organizations,” as amended.
- NIST Special Publication 800-53A-4, “Assessing Security and Privacy Controls in Federal Information Systems and Organizations: Building Effective Assessment Plans,” as amended.

- NIST Special Publication 800-161, "Supply Chain Risk Management Practices for Federal Information Systems and Organizations," as amended.
- NIST Special Publication 800-171, "Protecting Controlled Unclassified Information in Nonfederal Information Systems and Organizations," as amended.

11.3 Security and Privacy Requirements

FIPS 200, "Minimum Security Requirements for Federal Information and Information Systems," is a mandatory federal standard that defines the minimum-security requirements for federal information and information systems in seventeen security-related areas. Contractor systems supporting USCB must meet the minimum-security requirements using the security controls in accordance with NIST Special Publication 800-53, Revision 4 (as amended and hereafter described as NIST 800-53), and "Recommended Security and Privacy Controls for Federal Information Systems.

Data contained within all USCB computer systems are governed by Agency record disclosure and privacy regulations (13 C.F.R. part 102), IT Security regulations as well as other regulations, statutes and guidance, including the Privacy Act of 1974, as amended (5 U.S.C. § 552a). In addition, various Federal requirements obligate USCB to establish controls to limit access to Personally Identifiable Information (PII) and sensitive data, as defined below, to authorized personnel. These include OMB Circular A-130 and OMB Memoranda M-19-17, M-10-15, M-09-29, M-08-21, M-07-19, M-07-16, M-06-16, and M-06-15.

Contractors leveraging cloud solutions must utilize a Cloud Service Provider (CSP) with an existing Federal Risk and Authorization Management Program (FedRAMP) Joint Authorization Board (JAB) Provisional Authorization to Operate (ATO) or Agency ATO at all service models [Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS)].

To comply with the Federal standard, the USCB must determine the security category of the information and information system in accordance with FIPS 199, "Standards for Security Categorization of Federal Information and Information Systems", and then the contractor shall apply the appropriately tailored set of Low, Moderate, or High impact baseline security controls in NIST 800-53, as determined by the USCB.

The Contractor shall use USCB and NIST guidelines, Defense Information Security Agency (DISA) Security Technical Implementation Guides (STIGs), or industry guidelines in securing their systems.

11.4 Supply Chain Risk Management

The contractor shall, as part of their technical proposal, describe how Supply Chain Risk Management (SCRM) is incorporated into the acquisition of the underlying technology that composes their solutions. Descriptions shall at a minimum describe how the contractor's solutions satisfy the following regulatory components:

- The TAA of 1979, as amended.
- The Federal Acquisition Regulation (FAR) part 52.204-23 Representation Regarding Certain Telecommunications and Video Surveillance Services or Equipment.
- The Federal Acquisition Regulation (FAR) part 52.204-24 Representation Regarding Certain Telecommunications and Video Surveillance Services or Equipment.

- The Federal Acquisition Regulation (FAR) part 52.204-25 Representation Regarding Certain Telecommunications and Video Surveillance Services or Equipment.
- The Federal Acquisition Regulation (FAR) part 52.204-26 Representation Regarding Certain Telecommunications and Video Surveillance Services or Equipment.

The contractor's technical proposal shall describe how SCRM will be addressed and monitored throughout the lifecycle of the acquisition.

11.5 Essential Security Controls

All NIST 800-53 controls must be implemented as per the applicable FIPS 199 Low, Moderate, or High baseline. Controls, control enhancements, or parts of controls or enhancement may be implemented jointly between the Contractor and CSP as applicable. The following table identifies essential security controls from the respective baselines to highlight their importance and to understand the potential implementation costs. The Contractor shall make the proposed system and security architecture of the information system available to the Office of the Chief Information Officer for review and approval before commencement of system build.

Control ID	Control Title	Baseline	Implementation Guidance
AC-02	Account Management	L, M, H	The contractor shall perform an annual user recertification for all roles implemented within the information system.
AC-17 (3)	Remote Access | Managed Access Control Points	M, H	The information system routes privileged authentication traffic to external hosted infrastructures / applications through USCB's managed network access control points and are subject to the Trusted Internet Connections (TIC) and the U.S. Department of Homeland Security's (DHS') Einstein monitoring system.
AC-20	Use of External Information Systems	L, M, H	Physical access to the contractor's office areas that contain PII, and sensitive data shall be controlled to prevent unauthorized personnel from acquiring access to this data. The contractor shall not release USCB data outside of its facility, either orally or in written form, without the express written consent of the USCB CO/COR.
AU-02	Audit Events	L, M, H	Information systems shall implement audit configuration requirements including successful and unsuccessful account logon events, account management events, object access, policy change, privilege functions, process tracking, and system events. Web applications should log all admin activity, authentication checks, authorization checks, data deletions, data access, data changes, and permission changes. Web applications should log all admin activity, authentication checks, authorization checks, data deletions, data access, data changes, and permission changes.

Control ID	Control Title	Baseline	Implementation Guidance
AT-02/ AT-03	Security Awareness Training	L, M, H	All contractors with access to USCB's IT systems and/or PII and sensitive data shall complete the annual Computer Security Awareness Training (CSAT) and role-based training as necessary
CA-07	Continuous Monitoring	L, M, H	Information systems, including vendor owned / operated systems on behalf of the USCB, shall integrate with USCB implemented continuous monitoring, Continuous Diagnostics and Monitoring (CDM), and Security Operation Center (SOC) capabilities.
CA-08	Penetration Testing	L, M, H	The contractor shall support providing all physical and logical access necessary to the target system(s), in annual and ad hoc agency penetrations testing efforts.
CM-06	Configuration Settings	L, M, H	Information systems, including vendor owned / operated systems on behalf of the USCB, shall configure their systems in agreement with USCB policies and guidelines, DISA STIGs, NIST guidelines, or manufacturer guidelines as appropriate.
CP-07	Alternative Processing Site	M, H	FIPS 199 Moderate and High impact systems must implement processing across geographically disparate locations to ensure fault tolerance. Infrastructure as a Service architecture must implement a multi-region strategy in accordance with agency DR/COOP architecture.
CP-08	Telecom Services	M, H	FIPS 199 Moderate and High impact information systems must identify an alternate agency provided telecom service to support resumption of operations when the primary telecommunications capabilities are unavailable at either the primary or alternate processing or storage sites.
IA-02 (1)	Identification and Authentication (Organizational Users) | Network Access to Privileged Accounts	L, M, H	All information systems shall implement multi-factor authentication for privileged accounts.
IA-02 (2)	Identification and Authentication (Organizational Users) | Network Access to Non-	L, M, H	FIPS 199 Moderate and High impact information systems must implement multi-factor authentication for non-privileged accounts.

Control ID	Control Title	Baseline	Implementation Guidance
	Privileged Accounts		
IA-02 (12)	Identification and Authentication | Acceptance of PIV Credentials	L, M, H	Information systems with an e-authentication assurance level of 3 or above, used by federal employees or contractors must accept federal Personal Identity Verification (PIV) cards and verify them in accordance with guidance in OMB M-11-33.
IA-07	Cryptographic Module Authentication	L, M, H	The information system shall implement FIPS 140-2 validated encryption modules for authentication functions. Reference: http://csrc.nist.gov/groups/STM/cmvp/documents/140-1/1401vend.htm
IR-04	Incident Handling	L, M, H	The Contractor shall cooperate with and support agency incident response activities, including, providing logical and physical access to compute resources and media for investigative, examination, or forensic purposes.
IR-06	Incident Reporting	L, M, H	The contractor shall report all suspected security incidents and privacy breaches to the USCB Security Operations Center within 1 hour and in accordance with DHS/CISA reporting requirements.
MP-04	Media Storage	M, H	Digital media including magnetic tapes, external/removable hard drives, flash/thumb drives, diskettes, compact disks and digital video disks shall be encrypted using a FIPS 140-2 validated encryption module.
MP-05	Media Transport	M, H	Digital media including magnetic tapes, external/removable hard drives, flash/thumb drives and digital video disks shall be encrypted using a FIPS 140-2 validated encryption module during transport outside of controlled areas.
PL-02	System Security Plan	L, M, H	In accordance with the agency SDLC processes, the contractor must support the ISSO in developing a final System Security Plan (SSP) within thirty (30) calendar days from contract award. The SSP must be updated at least annually. The SSP will be reviewed by the COR or designated technical point of contact. The SSP must document administrative, technical, and physical security measures at the contractor’s computer facility to protect PII and sensitive data from unauthorized disclosure, alteration, or misuse; prevent unauthorized access to the contractor’s computer system; and protect the availability of data and services to USCB. All

Control ID	Control Title	Baseline	Implementation Guidance
			controls and enhancements must be described in detail, focusing on how each control or enhancement is implemented. All technical controls must be described for all technologies included in the system boundary
PL-08	Information Security Architecture	M, H	All information system security architectures must be reviewed and approved by the Office of the Chief Information Officer prior to development or implementation.
PS-03	Personnel Screening	L, M, H	[Applies to acquisitions where the contract personnel are provided direct logical or physical access to agency data, compute resources, or offices.] Pursuant to Federal Acquisition Regulation (FAR) clause 52.204-9, incorporated into this solicitation and the resulting contract, each contractor and subcontractor must comply with Agency Personal Identity Verification (PIV) procedures. These procedures must be followed when the contractor and/or subcontractor will have unescorted physical access to a federally controlled facility, and/or access to a federally controlled information system (including computer systems, networks, or information technology infrastructure).
PS-04	Personnel Termination	L, M, H	[Applies to acquisitions where the contract personnel are provided direct logical or physical access to agency data, compute resources, or offices.] The Contractor shall provide weekly in electronic format to the CO and COR a current roster of individuals with direct logical or physical access to agency data, compute resources, or offices. Roster shall indicate all changes (additions and departures).
RA-05	Vulnerability Scanning	L, M, H	All information systems must integrate with agency provided vulnerability scanning capabilities to ensure applicable scans are conducted and mitigated in accordance with agency and departmental requirements and timelines.
SA-04 / SA-04 (1)	Acquisition Process | Functional Properties of Security Controls	L, M, H	When applicable, all external service providers (contractor hosted, contractor managed, proprietary, IaaS, PaaS, SaaS, resellers, etc.) shall include as part of their technical proposal attestations of Service Level Agreements (SLAs) for the following key areas:

Control ID	Control Title	Baseline	Implementation Guidance
			○ Response to Cybersecurity incidents ○ Availability of service ○ Data retention ○ Encryption of data at rest ○ Telecommunication service redundancy ○ Logical and Physical Data Access Limited to Continental United States SLA commitments and/or attestations must either be included in the offeror's technical proposal or be provided by reference to publicly available Internet resources.
SC-08 / SC-08 (1)	Transmission Confidentiality and Integration	M, H	Implemented encryption algorithms and cryptographic modules shall be FIPS-approved and FIPS 140-2 validated, respectively. ○ Digital signature encryption algorithms - Reference: http://csrc.nist.gov/groups/ST/toolkit/digital_signatures.html#Approved ○ Block cypher encryption algorithms - Reference: Block Cipher Techniques | CSRC (nist.gov) https://csrc.nist.gov/projects/block-cipher-techniques ○ Secure hashing algorithms – Reference: Hash Functions | CSRC (nist.gov) https://csrc.nist.gov/projects/hash-functions Internet-facing systems shall enforce HTTPS and implement HTTP Strict Transport Security (HSTS).
SC-13	Cryptographic Protection | FIPS Validated Cryptography	L, M, H	Implemented encryption algorithms and cryptographic modules shall be FIPS-approved and FIPS 140-2 validated, respectively. ○ Digital signature encryption algorithms - Reference: http://csrc.nist.gov/groups/ST/toolkit/digital_signatures.html#Approved ○ Block cypher encryption algorithms - Reference: Block Cipher Techniques | CSRC (nist.gov) https://csrc.nist.gov/projects/block-cipher-techniques ○ Secure hashing algorithms – Reference: Hash Functions | CSRC (nist.gov) https://csrc.nist.gov/projects/hash-functions
SC-22	Architecture and Provisioning for Name / Address	L, M, H	Information systems shall be Domain Name System Security Extensions (DNSSEC) compliant as per OMB Memorandum, M-08-23, which requires all Federal Government departments and agencies that have

Control ID	Control Title	Baseline	Implementation Guidance
	Resolution Service		registered and are operating second level .gov to be DNSSEC.
SC-28 (1)	Protection of Information at Rest | Cryptographic Protection	Required for Systems with PII Only	System bearing PII must implement protect information at rest. At a minimum, fields bearing PII data must be encrypted with field level encryption. Encryption algorithms shall be FIPS-approved; implemented encryption modules shall be FIPS 140-2 validated
SI-02	Flaw Remediation	L, M, H	All systems must establish consistent flaw remediation (patch) processes. High and Critical risk findings must be remediated prior to go-live. Post go-live, all critical [CVSS Base Score = 10] vulnerabilities identified must be mitigated within 15 calendar days, high [CVSS Base Score ≥ 7.0] vulnerabilities identified must be mitigated within 30 calendar days and all other [CVSS Base Score < 7.0 vulnerabilities mitigated within 90 calendar days.
SI-03	Malicious Code Protection	L, M, H	Information systems, including vendor owned / operated systems on behalf of the USCB, shall implement agency enterprise anti-malware capabilities for all assets.
SI-04	Information System Monitoring	L, M, H	Information systems, including vendor owned / operated systems on behalf of the USCB, shall integrate with USCB implemented continuous monitoring, Continuous Diagnostics and Monitoring (CDM), and Security Operation Center (SOC) capabilities.
SI-10	Information Input Validation	M, H	All systems accepting input from end users must validate the input in accordance with the OWASP Top 10 Web Application Security Vulnerabilities and best practices.
SI-12	Information Handling and Retention	L, M, H	When applicable, all external service providers (contractor hosted, contractor managed, proprietary, IaaS, PaaS, SaaS, etc.) must describe, as part of their technical proposal, the means for ensuring that agency data stored and processed within the offeror's solution is portable, interoperable, and owned by the agency. Specifically, the offeror shall describe in detail the mechanism for exporting all agency data and meta-data into a Comma Separated Values (CSV), eXtensible Markup Language (XML), or equivalent structured format. This mechanism shall be available to agency administrators as a routine practice, shall not have proprietary software nor Application Programming Interface (API) dependencies, and shall result in no additional cost to the agency, however often the mechanism is used.

11.6 Assessment and Authorization (A&A) Activities

The implementation of a new Federal Government IT system requires a formal approval process known as Assessment and Authorization (A&A). NIST SP 800-37-2 provides guidelines for performing the A&A process. The Contractor system/application must have a valid assessment and authorization, known as an Authority to Operate (ATO) (signed by the Federal government) before going into operation and processing USCB information. The failure to obtain and maintain a valid ATO may result in the termination of the contract. The system must enter the Ongoing Authorization phase and have an ongoing authorization decision signed by the Authorizing official annually or at the discretion of the Authorizing Official when there is a significant change to the system's security posture. All NIST 800-53-4 controls must be tested/assessed as part of Information System Continuous Monitoring and as defined by USCB policy.

Assessing the System

a. The Contractor shall comply with, and support Assessment and Authorization (A&A) requirements as mandated by Federal laws and policies, including making available any documentation, physical access, and logical access needed to support this requirement. The Level of Effort for the A&A is based on the System's NIST Federal Information Processing Standard (FIPS) Publication 199 categorization. The contractor shall work with the ISSO to create, maintain and update the following A&A documentation:
   - System Security Plan (SSP) completed in agreement with NIST SP 800-18-1 (or successor document), "Guide for Developing Security Plans for Federal Information Systems". The SSP shall include as appendices required policies and procedures across 18 control families mandated per FIPS 200, Rules of Behavior, and Interconnection Agreements (in agreement with NIST Special Publication 800-47, "Security Guide for Interconnecting Information Technology Systems").
   - Contingency Plan and test Report completed in agreement with NIST Special Publication 800-34.
   - Privacy Threshold Analysis/Privacy Impact Assessment (PTA/PIA)
   - Configuration Management Plan (CMP).

b. Information systems must be entered into the ongoing authorization program and must be assessed whenever there is a significant change to the system's security posture and updated annually in accordance with NIST Special Publication 800-37-2.

c. At the Moderate impact level and higher, the contractor or Government (as determined in the contract) will be responsible for providing an independent Security Assessment/Risk Assessment in accordance with USCB Policy.

d. If the Government is responsible for providing a Security Assessment/Risk Assessment, the Contractor shall allow USCB employees (or USCB-designated third-party contractors) to conduct A&A activities to include control reviews in accordance with NIST 800-53/NIST 800-53A. Review activities include, operating system vulnerability scanning, web application scanning, and database scanning of applicable systems that support the processing, transportation, storage, or security of USCB information. This includes the general support system infrastructure.

e. Identified gaps between required 800-53 controls and the contractor's implementation as documented in the Security Assessment/Risk Assessment report shall be tracked for mitigation as a Plan of Action and Milestones (POAM) item within USCB's Governance Risk and Compliance tool implementation. Depending on the severity of the gaps, the Government may require them to be remediated before an Authorization to Operate is issued.

f. The Contractor is responsible for mitigating all security risks found during A&A and continuous monitoring activities. All critical-risk vulnerabilities must be mitigated within 15 days, high-risk vulnerabilities must be mitigated within 30 days, and all moderate risk vulnerabilities must be mitigated within 90 days from the date vulnerabilities are formally identified. The Government will determine the risk rating of vulnerabilities.

Authorization of the System

a. Upon receipt of the Security Authorization Package, the Authorizing Official (AO), in coordination with the USCB Office of the CIO, System Owner (SO), Information System Security Manager (ISSM), and Information System Security Officer (ISSO) will render an authorization decision to:
    - Authorize system operation w/out any restrictions or limitations on its operation to include Title 13 and or Title 26 data.
    - Authorize system operation w/ restriction or limitation on its operation, or;
    - Not authorize for operation.

b. The Contractor shall provide access to the Federal Government, or their designee acting as their agent, when requested, in Order to verify compliance with the requirements for an Information Technology security program, and in response to security incidents. At its option, the Government may choose to conduct on site surveys. The Contractor shall make appropriate personnel available for interviews and documentation during this review. If documentation is considered proprietary or sensitive, these documents may be reviewed on-site under the hosting Contractor's supervision.

11.7 Continuous Monitoring Deliverables and Schedule

Maintenance of the security authorization to operate will be through continuous monitoring of security controls of the contractor's system and its environment of operation to determine if the security controls in the information system continue to be effective over time in light of changes that occur in the system and environment. Through continuous monitoring, security controls and supporting deliverables are updated and submitted to USCB in accordance with the system's information system continuous monitoring plan. The submitted deliverables (or lack thereof) provide a current understanding of the security state and risk posture of the information systems. They allow USCB AOs to make credible risk-based decisions regarding the continued operations of the information systems and initiate appropriate responses as needed when changes occur. The contractor is required to provide deliverables identified through continuous monitoring to the CO/COR during the performance of the contract.

12.0 Data Rights

The government requires unlimited rights in any material first produced in the performance of this BPA. In addition, for any material first produced in the performance of this BPA, the materials may be shared with other agencies or Contractors during the period of performance of this BPA, or after its termination. For any subcontractors or teaming partners, Contractors shall ensure at proposal submission that the subcontractors and/or partners are willing to provide the data rights required under this BPA.

All documents and materials, to include the source code of any software produced under this BPA, shall be government owned and the property of the government with all rights and privileges of ownership/copyright belonging exclusively to the government. These documents and materials may not be used or sold by the Contractor without written permission from the CO. All materials supplied to the Government shall be the sole property of the government and may not be used for any other purpose. This right does not abrogate any other government rights.

13.0 Conflict of Interest

Contractor and subcontract personnel performing work under this Order may receive, have access to, or participate in the development of proprietary or source selection information (e.g., cost or pricing information, budget information or analyses, specifications, or work statements, etc.), or perform evaluation services which may create a current or subsequent Organizational Conflict of Interests (OCI) as defined in FAR Subpart 9.5. The Contractor shall notify the CO immediately whenever he/she becomes aware that such access or participation may result in any actual or potential OCI and may merit the submittal of a plan to the CO to avoid or mitigate any such OCI. This mitigation plan shall be determined to be acceptable solely at the discretion of the CO. In the event the CO unilaterally determines that any such OCI cannot be satisfactorily avoided or mitigated, the Contracting Officer may employ other remedies as he or she deems necessary, including prohibiting the Contractor from participation in subsequent contracted requirements which may be affected by the OCI. The USCB reserves the right to take all responsive measures (including Contractor termination) available under applicable laws and regulations.

14.0 Glossary

Glossary Term	Description
508 Compliance	Section 508 of the Rehabilitation Act ("Section 508") requires that when federal agencies develop, procure, maintain, or use electronic and information technology, the agencies shall ensure that technology allows: a. Federal employees with disabilities to have access to and use of information and data that is comparable to that by federal employees who are not individuals with disabilities, unless an undue burden would be imposed on the agency. b. Individuals with disabilities, who are members of the public seeking information or services from a federal agency, to have access to and use of information and data that is comparable to that provided to the public who are not individuals with disabilities. Moreover, Section 508 regulations that pertain specifically to web sites (36 CFR 1194.22) are adopted as a DOC standard. For an overview of the regulations, see www.section508.gov
Integrated Master Schedule (IMS)	IMS provides the program team with a program execution roadmap of meaningful progress and realistic forecasts against a resource-loaded performance measurement baseline. The primary purpose of any IMS is to help the Program Manager and the Program Team optimize the overall execution strategy of a program, coordinate workflows, and assist in the decision-making processes to mitigate risks and resolve challenges on a day-to-day basis.
Integration and Implementation Plans (IIP)	The framework for integrating Information Technology (IT) solutions that make up the GEO programs. It establishes Operational Delivery (OD) and Readiness Reviews.
independent Verification and Validation (IV&V)	A comprehensive review, analysis, and testing (software and/or hardware) performed by an objective third party to confirm (i.e., verify) that the requirements are correctly defined, and to confirm (i.e., validate) that the system correctly implements the required functionality and security requirements.
Operational Delivery (OD)	The technical, logistical, and in-field work that must be coordinated to be deployed within the same timeframe and often using shared system functionality. The focus is on implementing and testing the operation and solution delivery together. Operational Delivery teams, in collaboration with project teams and stakeholders, will define the scope of each OD.
ODMP	Identifies all processes and procedures used to release and deploy systems in accordance with the GEO IIP
Runbooks	A comprehensive set of steps detailing system and user interactions when releasing new software or features into a production environment. The runbook is inclusive of selecting the users to participate, coordination with the stakeholders and solution providers involved, monitoring system performance and providing the venue for feedback based on predefined success criteria. The goal is to mitigate risks and provide for a smooth transition to a wider audience.
Readiness Reviews	Planned, documented, management methodology used to evaluate the preparedness of a set of activities that provides visible, objective, and independent evidence that prerequisites have been satisfied, and administrative and technical procedures have been reviewed.
Service Provider	A team responsible for providing specific services or resources, such as database administration, testing, or IT infrastructure.
Solution Provider	A team within Census that builds systems that makes up the Systems of Systems for Decennial 2030.

Glossary Term	Description
Systems Engineering Management Plan (SEMP)	The plan that provides the specifics of the technical effort and describes what technical processes will be used, how the processes will be applied using appropriate activities, how the project will be organized to accomplish the activities, information flow within the organization, decision making structure, and the resources required for accomplishing the activities. The process activities are driven by the critical events during any phase of a life cycle (including operations) that set the objectives and work product outputs of the processes and how the processes are integrated
Test and Evaluation Master Plan (TEMP)	The plan that describes the overall structure, objectives, resources, and schedule of the test and evaluation (T&E) program for a system or program. It covers the life-cycle phases from initiation to post deployment and supports major program decisions. It also identifies the tasks, activities, methodologies, responsibilities, and reporting for test activities. The TEMP is the primary test management document and the focal point of the development, review, and approval process for detailed T&E plans. The TEMP is developed and approved in accordance with the relevant policies and guidelines, such as the DoD Instruction and the FAA Handbook.

3 Acronyms

Acronym	Description
ART	Agile Release Train
CCB	Change Control Boards
Continuous Delivery (CD)	Continuous Delivery
Continuous Deployment	Continuous Deployment
CMP	Contract Management Plan
COP	Community of Practice
CSP	Cloud Service Provider
CI	Continuous Integration
CPR	Contractor Performance Review
EWG	Enterprise Working Group
eSDLC	Enterprise Software Development Life Cycle
EVMS	Earned Value Management Systems
GFE	Government Furnished Equipment
IaC	Infrastructure as Code
IIP	Integration and Implementation Plans
IMS	The Integrated Master Schedule
IMP	Integrated Master Plan
MCASR	Monthly Contract Administration Status Report
MOA	Memorandum of Agreement
OIS	The Office of Information Security
OD	Operational Delivery
ORRs	Operational Readiness Reviews
PI	Program Increment
PM	Project Management
PMR	Program Management Reviews
PRRs	Production Readiness Reviews
QASP	Quality Assurance Surveillance Plan
QCP	Quality Control Plan

Acronym	Description
SaC	Security as Code
SAFe®	Scaled Agile Framework
SEMP	Systems Engineering Management Plan
SOPs	Standard Operating Procedures
SoS	System of Systems
SOW	Statement of Work
TEMP	Testing and Evaluation Management Plans
TIP	Transition-In Plan
TOP	Transition-Out Plan
TRB	Technical Review Board
TRR	Technical Review Report
V&V	Verification and Validation
WBS	Work Breakdown Structure
ZTA	Zero Trust Architecture